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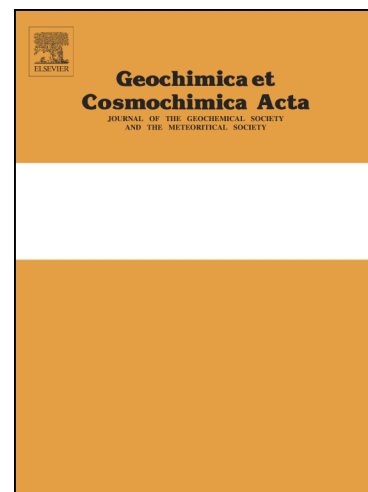
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**Microchemical and Sulfur Isotope Variation in Pyrite-Marcasite: A
Proxy for Low-Temperature Ore-forming Processes in Submarine
Hydrothermal Systems**

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Abstract

Pyrite-marcasite assemblages, formed at steep physicochemical gradients in submarine hydrothermal systems, serve as valuable records of fluid evolution and ore-forming processes. However, gaps remain in our understanding of the detailed pathways of pyrite-marcasite formation and their dynamics of trace element enrichment in seafloor massive sulfide deposits. Here, we utilize in-situ trace element and sulfur isotope analysis to systematically explore the microchemical variations within pyrite-marcasite assemblages from the basalt-hosted Xunmei and Niaochao vent fields on the slow- and fast-spreading ridges, respectively.

Diverse pyrite-marcasite pairs were identified across sulfide chimneys based on grain size, texture, and mineral sequence, each displaying distinct microchemical characteristics. Colloform marcasite-pyrite intergrowths, formed at the outermost chimney walls, exhibit metal enrichment and isotopically light sulfur in marcasite ($\delta^{34}\text{S}$ values as low as 0.72‰) compared to pyrite, attributed to high surface-to-volume ratios and sulfur isotope exchange during the rapid crystallization of marcasite. In the transition layers, the assemblage is formed through the replacement of subhedral pyrite with anhedral marcasite, resulting in the redistribution of metals (e.g., As, Ag, Au, Co, Pb) and the creation of metal-rich zones. Variations in sulfur isotopes within this pair are linked to the reduction of anhydrite by hot fluids. In contrast, the dense pyrite-marcasite assemblages, formed through recrystallization processes, show metal depletion. Their distinctly high $\delta^{34}\text{S}$ values (mean: 4.61‰, $n = 11$) are caused by sulfate reduction in the subseafloor hydrothermal stockwork. During seawater alteration, the marcasite pseudomorphs after pyrrhotite, cemented by anhedral pyrite in the middle layer, are enriched in hydrothermal metals such as Sb, Co, Zn, Cu, and Au, likely due to the adsorption reaction. Meanwhile, the secondary marcasite exhibits elevated $\delta^{34}\text{S}$ values (up to 6.41‰) as a result of anhydrite reduction via pyrrhotite oxidation. These microchemical patterns of pyrite-marcasite assemblages provide critical insights into the redistribution and re-enrichment of metals during low-temperature mineralization and replacement in submarine hydrothermal systems.

Keywords: *Pyrite-marcasite, Trace elements, Sulfur isotopes, Metal redistribution, Replacement and recrystallization, Submarine hydrothermal systems*

1. Introduction

Submarine hydrothermal circulation plays a critical role in triggering chemical exchange and heat transfer between the lithosphere and the oceans (Haymon, 1983; Alt, 1995). Seafloor massive sulfide (SMS) deposits, as its key outputs, are enriched in Cu, Co, Ni, and Au, offering a promising resource for global metal demands (Petersen et al., 2018). Trace element variations in sulfides provide valuable insights into the mineralization environment, metal sources, and ore-forming processes of submarine hydrothermal systems (e.g., Wohlgemuth-Ueberwasser et al., 2015; Falkenberg et al., 2021; Dang et al., 2023). Pyrite and its dimorph (marcasite) are the most prevalent sulfides and principal metal carriers within SMS deposits (Graham et al., 1988; Keith et al., 2016a, b; Grant et al., 2018; Choi et al., 2021). Pyrite stabilized at a wide temperature range (up to 400°C) and under acidic-to-neutral conditions, whereas marcasite is only stable at lower temperatures (below 240°C) and in low-pH environments (typically pH < 5) (Murowchick and Barnes, 1986; Murowchick, 1992). Marcasite is metastable and often transitions to pyrite due to physicochemical changes and replacement processes (Murowchick, 1992; Yao et al., 2020). Therefore, pyrite-marcasite assemblages and intergrowths serve as markers of distinct physicochemical gradients and fluid evolution (Graham et al., 1988; Maslennikov et al., 2009).

In sulfide chimney margins, colloform pyrite-marcasite intergrowths, formed during the hydrothermal fluid-seawater interaction, are enriched in Au, As, Sb, Pb, Tl, Mn, Ni, and Ba (Maslennikov et al., 2009). In contrast, the recrystallization of these assemblages due to low-temperature hydrothermal overprinting tends to deplete most elements, except for Mn, Ni, Se, and Bi (Falkenberg et al., 2021). In-situ trace-element results of sulfides from the Trans-Atlantic Geotraverse (TAG) vent field exemplify that anhedral marcasite, formed at lower temperatures (< 240°C), is characterized by high concentrations of Zn, As, Sb, Pb, and Tl, whereas euhedral pyrite, precipitated at higher temperatures (>240°C), is more enriched in Cu, Co, Se, and Bi (Grant et al., 2018). In shallow marine hydrothermal systems (i.e., Milos and Calypso vents), marcasite shows increased concentrations of volatile elements such as Hg and Bi due to subseafloor fluid boiling (Nestmeyer et al., 2021). These findings suggest that the chemical signatures of pyrite and marcasite can effectively reflect fluid evolution and ore-forming processes, particularly low-temperature hydrothermal mineralization. However, the mechanisms of pyrite-marcasite formation and the partitioning of metals between these minerals remain elusive in submarine hydrothermal systems.

Sulfur isotopes provide a robust tool for tracing fluid sources and environmental changes during the precipitation of pyrite and marcasite (e.g., Herzig et al., 1998; Rouxel et al., 2004, 2008; Falkenberg et al., 2022). In sediment-starved mid-ocean ridge systems, sulfur is primarily derived from igneous rock leaching ($\delta^{34}\text{S}$: $0.1 \pm 0.5\text{‰}$; Sakai et al., 1984) and seawater sulfate reduction ($\delta^{34}\text{S}$: $\sim 20.9\text{‰}$; Rees et al., 1978). For example, interactions between hydrothermal fluids and seawater elevate $\delta^{34}\text{S}$ values of sulfides within mounds and chimneys (Bluth and Ohmoto, 1988; Meng et al., 2020, 2024). The temperature heterogeneity across sulfide chimneys can induce sulfur isotopic variations of sulfides (Syverson et al., 2015; Wang et al., 2020). Additionally, significant isotopic fractionation would occur during magmatic degassing, bacterial sulfate reduction, and fluid boiling (Herzig et al., 1998; Alt et al., 2007; Ding et al., 2021; Falkenberg et al., 2021).

Basalt-hosted hydrothermal systems, including the Xunmei and Niaochao vent

fields located on the Southern Mid-Atlantic Ridge (SMAR) and the East Pacific Rise (EPR), respectively, are characterized by diverse pyrite-marcasite assemblages (Meng et al., 2020; Dang et al., 2023). Compared to ultramafic-hosted systems, mafic-hosted SMS deposits produce more marcasite (Fouquet et al., 2010; Wang et al., 2014). To gain a comprehensive understanding of the mineralogical and geochemical characteristics of pyrite-marcasite pairs in submarine vent systems along mid-ocean ridges, it is essential to select sulfide samples from mafic-hosted fields on the slow- and fast-spreading ridges (i.e., SMAR and EPR).

Considering the intricate intergrowth and replacement textures of pyrite and marcasite within SMS deposits, we employed laser ablation inductively coupled plasma mass spectrometry (LA-ICP-MS) and laser ablation multicollector inductively coupled plasma mass spectrometry (LA-MC-ICP-MS) to analyze in-situ trace elements and sulfur isotopes, thereby elucidating their microchemical characteristics. Our findings highlight that the variations in trace element and sulfur isotope compositions between paired pyrite and marcasite, influenced by rapid crystallization during fluid-seawater mixing, evolving fluid conditions, and processes of replacement, recrystallization, and mineral transformation, offer a new window into environmental changes and mineralization processes of submarine hydrothermal systems.

2. Geological background

2.1. *Xunmei vent field*

The Xunmei hydrothermal field is located between 25°40'S and 26°35'S along the Southern Mid-Atlantic Ridge (SMAR), a slow-spreading ridge characterized by an uneven spreading rate—approximately 19.3 mm/yr to the west and 16.3 mm/yr to the east (Carbotte et al., 1991). First identified and explored during the Chinese DY22nd cruise in 2011 aboard the research vessel *Dayang Yihao* (Li et al., 2014), the field is located at a water depth of approximately 2600 m, about 37 kilometers southeast of the Fangzhou seamount (www.gebco.net). The basement rocks primarily consist of mid-ocean ridge basalts (MORBs), with minimal sedimentary cover (Niu and Rodey, 1994). Extensive collapsed sulfide chimneys and massive sulfide deposits have been sampled from slopes and low-lying areas, but no fluid samples were collected. The sulfide chimneys and massive sulfides at the Xunmei vent field comprise mostly pyrite, marcasite, and chalcopyrite (Fan et al., 2022; Dang et al., 2023), which were significantly oxidized by seawater and Si-rich fluids (Meng et al., 2021).

Sulfide chimney samples (e.g., 46II-S23-TVG17 and 46II-S33-TVG26) were collected from inactive black smokers using a television grab (TV-grab) during the Chinese DY46th cruise in 2017 (Fig. 1a, b). These samples were obtained at coordinates 13°51.30'W, 26°1.68'S and 13°51.17'W, 26°1.05'S, at seawater depths of approximately 2580 m and 2549 m, respectively.

2.2. *Niaochao vent field*

The Niaochao hydrothermal field is located along the East Pacific Rise (EPR) near 1–2°S, a region characterized by fast to ultrafast spreading rates of 80–200 mm/year (Holler and Marchig, 1990). It was first discovered near the Galápagos hotspot at a depth of approximately 2900 m (Tao et al., 2008). The seafloor topography of the Niaochao vent field features a relatively flat central region, bordered by low-lying terrain gently sloping towards the flanks (Zeng et al., 2015b). The basement is dominated by fresh pillow basalts, with minimal sediment cover (Macdonald and Fox, 1983). Extensive massive sulfide deposits and sulfide chimneys, primarily comprising pyrite, chalcopyrite, and sphalerite, were discovered and sampled during 2008–2015 expeditions (Zeng et al., 2015b; Meng et al., 2024). These sulfides are enriched in Fe, Cu, Zn, and other metals, suggesting that the Niaochao field has the crucial economic potential of polymetallic sulfide resources (Zeng et al., 2015a, b; Meng et al., 2020).

For this study, sulfide fragments—remnants of collapsed sulfide chimneys (e.g., 20III-S4-TVG1-a and -b)—were retrieved using a TV-grab during the DY20th cruise in 2008 (Fig. 1a, c). The samples are situated at coordinates 1°22.13'S; 102°37.36'W, with a seawater depth of approximately 2750 m.

3. Methods

3.1. *Mineralogical studies*

Mineral phases and textural relationships were examined using optical microscopy and field-emission scanning electron microscopy (SEM) (Zeiss Ultra 55) at the Second Institute of Oceanography, Ministry of Natural Resources (SIO, MNR), Hangzhou, China. The SEM, equipped with energy dispersive spectroscopy (EDS, Oxford INCA

MAX 20), was operated in backscattered electron (BSE) and secondary electron modes, with an accelerating voltage of 15 kV and a beam current of 16–22 nA. Under optical microscopy, marcasite is identified by its pewter color and relatively small crystal size, in contrast to pyrite, which displays a brassy color and relatively large crystal size (Figs. 2, 3). BSE imaging further reveals that pyrite exhibits a higher optical intensity than marcasite (Fig. 3).

3.2. Major element composition

Major element compositions of sulfides were analyzed by electron probe microanalysis (EPMA) using a JEOL JXA-8100 EPMA at SIO, MNR. The operating conditions were as follows: beam current of 10 nA, beam size of 1–5 μm , and accelerating voltage of 20 kV. Reference standards, including natural and synthetic minerals, were sourced from USA SPI Supplies. The detailed data of sulfides and reference standards are provided in Table S1.

3.3. Trace element composition

3.3.1. Micro-XRF microscopy

Transverse sections were analyzed using desktop micro-beam X-ray fluorescence (Micro-XRF) microscopy at China University of Petroleum (East China), Qingdao, China, with a Bruker M4 TornadoTM system. The instrument is equipped with a rhodium target X-ray tube operating at 50 kV and 800 nA, and two XFlash[®] silicon drift X-ray detectors (Flude et al., 2017). Fluorescence radiation was excited by the Rh anode side-window X-ray tube, with a polycapillary lens providing a 40- μm spot size on a 20- μm raster and a dwell time of 5 ms per pixel (Mao et al., 2017). To minimize errors from beam instability, the X-ray tube was pre-heated for at least 1.5 hours prior to analysis. Data were processed using the software M4 Tornado V1.6.614.0.

3.3.2. LA-ICP-MS trace elements

Laser ablation inductively coupled plasma mass spectrometry (LA-ICP-MS) was conducted using a Teledyne Cetac Technologies Analyte Excite laser ablation system coupled with an Agilent Technologies 7700x quadrupole ICP-MS at Nanjing FocuMS Technology Co. Ltd., Nanjing, China. The 193-nm ArF excimer laser, homogenized by a set of beam delivery systems, was focused on the sulfide surface with a fluence of 6 J/cm². Prior to LA-ICP-MS analysis, mineralogical observations were conducted to verify that the measured spot locations were free of mineral inclusions. Each spot analysis was preceded by a short pre-ablation to remove surface contaminants. The single spot was ablated using a 6 Hz repetition rate for 40 seconds, corresponding to 240 pulses. The total run time, including a 30-second pre-ablation blank analysis, was 70 seconds. The analytical spot size ranged from 32 to 44 μm , depending on the mineral grain size. Ablated material was transported via He carrier gas and mixed with Ar before entering the ICP mass spectrometer. A selection of elements, including ⁵⁷Fe, ³⁴S, ⁸²Se, ⁴⁷Ti, ⁵¹V, ⁵⁵Mn, ⁵⁹Co, ⁶⁰Ni, ⁶⁵Cu, ⁶⁶Zn, ⁷⁵As, ²⁰⁸Pb, ²⁰⁹Bi, ¹²¹Sb, ¹⁰⁷Ag, ¹⁹⁷Au, ¹¹⁵In, ¹¹⁸Sn, ²⁰⁵Tl, ¹²⁶Te, ⁹⁵Mo, ²⁴Mg and ²³⁸U was measured. The reference standards used include MASS-1 (Fe-Cu-Zn-S pressed pellet) and GSE-1G (synthetic basaltic glass) from the United States Geological Survey (Gao et al., 2015), with detailed quantification procedures described in Zhang et al. (2020). The analysis data were processed offline using the “ICPMS DataCal” software (Liu et al., 2008). Spots with

relatively smooth LA-ICP-MS down-hole spectra were retained. In cases where micro-inclusions were confined to a narrow segment within an individual down-hole spectrum, those intervals were excluded, and only clean regions free from noticeable inclusions were selected for further analysis. The relative standard deviations (RSD) for trace elements were < 10%, while for Au, the RSD was < 15%. The LA-ICP-MS results, along with their errors, detection limits, and standard data, are listed in [Tables S2 and S3](#).

3.3.3. LA-ICP-TOF-MS imaging

Laser ablation inductively coupled plasma time-of-flight mass spectrometry (LA-ICP-TOF-MS) is ideal for rapid multi-element imaging and analyzing very fast transient signals. The LA-ICP-TOF-MS system was equipped with a 193 nm ArF excimer laser ablation unit (ESI NWR Image GEO 193), coupled with an ICP-TOFMS (TOFWERK icpTOF R) at Shanghai Chemlab Laboratory, Shanghai, China. Helium was used as the carrier gas in the ablation cell, with a flow rate of ~600 ml/min. The sample was scanned using a laser spot diameter of ~20 μm , a repetition rate of ~20 Hz, and a laser fluence of ~7 J/cm², with a scan speed of ~4000 $\mu\text{m/s}$. The fast washout achieved with the ablation cell results in sharp ablation signals with minimal aerosol dispersion, while the washout time of 50 ms enables high sample throughput ([Burger et al., 2015](#)). The imaging data were processed using the software “Lolite”.

3.4. Sulfur isotope analysis

Sulfur isotope data for pyrite, marcasite, and chalcopyrite were analyzed using a Neptune Plus multicollector ICP-MS (Thermo Fisher Scientific, Bremen, Germany) coupled with a Geolas HD excimer ArF laser ablation system (Coherent, Göttingen, Germany) at Wuhan Sample Solution Analytical Technology Co. Ltd., Wuhan, China. Helium was used as a carrier gas, with argon as the make-up gas. To minimize down-hole fractionation effects, analysis parameters were set to a large spot size of ~44 μm , a slow pulse frequency of ~2 Hz, and a laser fluence of ~5 J/cm² ([Fu et al., 2016](#)). Only those areas in pyrite and chalcopyrite that were free of micro-inclusions were selected for analysis. Reference standards, including pyrite PPP-1 and chalcopyrite GBW07268, were selected to correct for natural pyrite and chalcopyrite isotopic analyses ([Fu et al., 2016](#)). The pyrite standard has been utilized to calibrate the sulfur isotope analysis of marcasite as reported by [Wu et al. \(2019\)](#). In-house reference materials, such as pyrrhotite SP-Po-01 ($\delta^{34}\text{S}_{\text{VCDT}} = 1.4\text{‰} \pm 0.4\text{‰}$) and chalcopyrite SP-CP-01 ($\delta^{34}\text{S}_{\text{VCDT}} = 5.45\text{‰} \pm 0.3\text{‰}$), were repeatedly analyzed as unknowns to verify calibration accuracy. Standard errors for PPP-1, GBW07268, SP-Po-01, and SP-CP-01 analyses were $\pm 0.09\text{‰}$ (2σ ; $n = 46$), $\pm 0.16\text{‰}$ (2σ ; $n = 12$), $\pm 0.16\text{‰}$ (2σ ; $n = 4$), and $\pm 0.22\text{‰}$ (2σ ; $n = 6$), respectively. All data, processed utilizing the “Iso-Compass” software, are detailed in [Table S4](#).

4. Results

4.1. Mineralogical characteristics

Sulfide chimneys and fragments from the Xunmei and Niaochao vent fields are predominantly composed of pyrite, marcasite, and chalcopyrite, with minor amounts of sphalerite. Elemental mapping of transverse sections of these sulfide samples reveals that Fe is ubiquitously distributed, Cu is enriched in the inner layers, and Zn exhibits a

sporadic pattern (Fig. 2). These observations indicate that pyrite and marcasite are the dominant minerals, with chalcopyrite comprising the interior conduit within the sulfide samples (Figs. 2, 3).

In the outermost layer of the Xunmei chimneys, laminated pyrite (I-Py1) was encased by colloform marcasite (I-Mrc1), forming marcasite-pyrite intergrowths (Fig. 3a, b), while at the transition layer, subhedral pyrite crystals (I-Py2), ranging from 50 to 300 μm in size, were enveloped by porous anhedral marcasite (I-Mrc2) (Fig. 3c, d). A minor amount of anhydrite (Anh) occurs within the pyrite interstices (refer to Meng et al., 2021). In the middle zone, marcasite pseudomorphs (I-Mrc3), characterized by a columnar morphology (Fig. 3e), were cemented by porous anhedral pyrite (I-Py3) and then altered by Fe-Si-rich fluids (Fig. 3f, Meng et al., 2021). Minor sphalerites (I-Sp) were replaced by chalcopyrite (I-Ccp1) and anhedral pyrite, forming a reaction-rim texture (Fig. 3g). Subsequently, euhedral pyrites (I-Py4) overgrew anhedral pyrite (I-Py3) (Fig. 3h). Granular I-Py4, with 50-300 μm in diameter, was surrounded by massive chalcopyrite (I-Ccp2) within the inner chimney conduit (Fig. 3i).

In the Niaochao vent field, colloform marcasite (II-Mrc1) rimmed the porous anhedral pyrite (II-Py1), forming intergrowths in the outermost layer of the chimney fragments (Fig. 3j). In the transition layer, the marcasite-pyrite assemblages arose from the replacement of earlier marcasite by later pyrite (Fig. 3k, l). The recrystallized massive pyrite (II-Py2) contains abundant small, oriented, and unoriented marcasite (II-Mrc2). II-Mrc2, exhibiting bladed crystals, occurs as polycrystalline, banded intergrowths with II-Py2 (Fig. 3m), as observed in the BSE image (Fig. 3n). In the inner layer, massive chalcopyrite (II-Ccp) replaced the earlier pyrite (Fig. 3o).

4.2. Geochemical characteristics

In the two vent fields, Fe and S contents in pyrite and marcasite (excluding I-Mrc3) are nearly consistent (Fig. S1). Trace element concentrations in pyrite and marcasite from the Xunmei vent field are generally higher than those from the Niaochao field, except for Sb, Mn, and Pb (Fig. 4). Selenium, In, and Sn concentrations in chalcopyrite show an opposite trend (Table S2). Notably, the grains of II-Mrc1 and II-Py1 in the Niaochao field are too small for LA-(MC)-ICP-MS analysis.

In the outermost zone of the Xunmei chimneys, I-Mrc1 exhibits higher median levels of Co, V, Zn, Mo, Mn, Tl, As, Au, Pb, and Cd compared to I-Py1 (Fig. 5a), while I-Py1 occasionally shows layers enriched with Fe, Co, Ag, and Pb (Fig. 6).

Moving toward the transition zone, I-Mrc2 is Co-rich with high concentrations of Mg, Tl, U, Ni, Ba, Cd, and Mn, while I-Py2 is As-rich with high levels of Au, Pb, Ag, Sb, and Zn (Figs. 5b, 7). Both exhibit relatively low Se/Tl and Co/Ni ratios compared to other pyrite and marcasite in the Xunmei field (Fig. 8a, b).

Similarly, within the transition layer of Niaochao chimney fragments, II-Mrc2 displays elevated median levels of Mo, Mn, U, Mg, Cu, Cd, V, Tl, Ni, and Co, while II-Py2 is enriched with As, Pb, Ag, Au, and Sb (Fig. 5c). EPMA mapping reveals that Fe, Cu, and Zn show no significant variations within pyrite-marcasite (Fig. S2).

Toward the middle zone of the Xunmei chimneys, I-Mrc3 is enriched in Sb, Co, Zn, Cd, Au, Cu, Ag, Se, and U compared to coexisting I-Py3 (Fig. 5d). Notably, Sb,

Co, Zn, Cd, and Au contents (median: 14.9, 6085, 3709, 6.63, and 0.84 ppm, respectively) are particularly high in I-Mrc3.

The euhedral I-Py4 crystals, which grew over I-Py3 in the interior zone of the Xunmei chimneys, exhibit the highest levels of Se, Se/Tl, and Co/Ni (median: 131 ppm, 502, and 2317, respectively; Fig. 8a, b) but the lowest concentrations of Zn, Au, Ag, Mn, Pb, Tl, Mg, and V (Fig. 4).

4.3. Sulfur isotope data

A total of sixty-three $\delta^{34}\text{S}$ measurements were conducted on pyrite, marcasite, and chalcopyrite from the two studied fields (Fig. 9a, b). At the Xunmei field, $\delta^{34}\text{S}$ values for pyrite range from 1.93 to 4.37‰ (mean: $3.48 \pm 0.65\text{‰}$, $n = 27$), marcasite from 0.72 to 6.41‰ (mean: $3.31 \pm 1.67\text{‰}$, $n = 13$), and chalcopyrite from 2.33 to 3.96‰ (mean: $3.38 \pm 0.62\text{‰}$, $n = 8$). In the Niaochao field, $\delta^{34}\text{S}$ values for pyrite vary from 4.08 to 5.09‰ (mean: $4.69 \pm 0.40\text{‰}$, $n = 5$), marcasite from 2.32 to 5.70‰ (mean: $4.54 \pm 1.28\text{‰}$, $n = 6$), and chalcopyrite from 6.37 to 7.57‰ (mean: $6.92 \pm 0.50\text{‰}$, $n = 4$) (Table S4). Marcasite displays a wider $\delta^{34}\text{S}$ range compared to pyrite, despite similar means.

In the Xunmei chimneys (Fig. 9c), $\delta^{34}\text{S}$ values show a progressive increase from I-Mrc1 (mean: $1.82 \pm 0.77\text{‰}$, $n = 4$) to I-Py1 (mean: $3.04 \pm 0.40\text{‰}$, $n = 3$) and I-Py2 (mean: $2.41 \pm 0.52\text{‰}$, $n = 3$), rising further in I-Mrc2 (mean: $4.26 \pm 1.99\text{‰}$, $n = 3$). In the middle zone, I-Py3, I-Mrc3, and I-Ccp1 display relatively high $\delta^{34}\text{S}$ values of $4.06 \pm 0.27\text{‰}$ ($n = 7$), $3.83 \pm 1.45\text{‰}$ ($n = 6$), and $3.86 \pm 0.10\text{‰}$ ($n = 3$), respectively. Moving toward the inner layer, a decrease in $\delta^{34}\text{S}$ values is observed in I-Py4 (mean: $3.52 \pm 0.49\text{‰}$, $n = 14$) and I-Ccp2 (mean: $3.09 \pm 0.62\text{‰}$, $n = 5$). In contrast, sulfides from the Niaochao field, including II-Mrc2, II-Py2, and II-Ccp, exhibit generally higher $\delta^{34}\text{S}$ values compared to those from the Xunmei field, with II-Ccp being particularly notable.

5. Discussion

5.1. Temporal-spatial relationship of pyrite and marcasite

Pyrite and marcasite commonly occur as cogenetic or coexisting assemblages within the sulfide chimneys and fragments at the Xunmei and Niaochao hydrothermal fields (Fig. 3). Four types of pyrite-marcasite (Py-Mrc) assemblages are identified based on their morphology, texture, grain size, and mineral sequence (Fig. 10).

In the outermost layer of the Xunmei chimneys, colloform marcasite (I-Mrc1) rimmed cogenetic pyrite (I-Py1), forming marcasite-pyrite intergrowths (Type 1 Py-Mrc, Figs. 3a, 10). These were typically precipitated at low temperatures ($< 150^\circ\text{C}$) when hot fluids were discharged into ambient seawater (Hannington et al., 1995; Xu et al., 2005). Similar marcasite-pyrite intergrowths also occur in the outermost layer of chimney fragments from the Niaochao vent field (Fig. 3j). Furthermore, such intergrowths are observed in the Urals volcanogenic massive sulfide (VMS) deposits (Maslennikov et al., 2009).

Toward the transition zone of sulfide chimneys, marcasite-pyrite assemblages formed through significant replacement processes (Type 2 Py-Mrc, Figs. 3c, 10). In the Xunmei chimneys, subhedral pyrite (I-Py2) was cemented by minor anhydrite, precipitating at temperatures exceeding 150°C (Tivey, 1998). Anhedral marcasite (I-

Mrc2), which precipitated at temperatures below 240°C (Murowchick and Barnes, 1986), partially replaced the earlier pyrites, forming a pyrite-marcasite pair (Fig. 3c). The precipitation of marcasite indicates sharp changes in physicochemical conditions, including temperature and redox states (Graham et al., 1988; Yao et al., 2020).

Similarly, in the transition layer at the Niaochao vent field, the dense marcasite-pyrite assemblage (Type 3 Py-Mrc, Figs. 3k, 10) results from the replacement of marcasite (II-Mrc2) by pyrite (II-Py2). The marcasite either recrystallized or transformed locally into pyrite (Murowchick, 1992), containing small blade-like marcasite inclusions (Fig. 3k-n). The marcasite is considered a remnant of early low-temperature ore-forming processes, commonly observed in the TAG vent field (Knott et al., 1998; Grant et al., 2018).

In the middle zone of the Xunmei chimneys (indicated by yellow dotted lines in Fig. 2a1-3), columnar marcasite (I-Mrc3) and coexisting anhedral pyrites (I-Py3) are present (Type 4 Py-Mrc, Figs. 3e, 10). The marcasite pseudomorph was formed through the replacement of primary pyrrhotite by low-temperature (below 220°C) and oxidized fluids, which were generated by the mixing of hot fluids with seawater (Qian et al., 2011; Choi et al., 2021). The secondary marcasite was later altered by Si-rich fluids (Fig. 3f), leading to the destruction of its original textures (Meng et al., 2021).

As hydrothermal activity intensified, anhedral pyrite (I-Py3) was overgrown by euhedral pyrite (I-Py4) (Fig. 3h; Grant et al., 2018). The Se/Tl and Co/Ni ratios are indicators of temperature (Maslennikov et al., 2009; Keith et al., 2016b), and the highest values in I-Py4 (Fig. 8a, b) suggest the fluid temperature is elevated. This is evidenced by the calculated temperatures (259 to 299°C with a mean of 282°C, Table S1) based on sphalerite Fe/Zn ratios (Keith et al., 2014). The precipitation of minor chalcopyrite (I-Ccp1) filling among I-Py4 crystals also indicates increased precipitation temperatures (~300°C, George et al., 2018; Lehrmann et al., 2022). As the chimney matured (~350°C), massive chalcopyrite (e.g., I-Ccp2 and II-Ccp, Fig. 3i, o) replaced earlier marcasite and pyrite to form the central conduits of chimneys (Haymon, 1983).

5.2. Metal partitioning between pyrite and marcasite

5.2.1. Rapid crystallization of marcasite-pyrite

The colloform marcasite-pyrite intergrowths (Type 1 Py-Mrc, specifically I-Mrc1 and I-Py1), observed in the outermost layer of the Xunmei chimneys (Fig. 3a, b), were precipitated at low temperatures during rapid fluid-seawater mixing (Graham et al., 1988; Kim et al., 2006). I-Mrc1 is notably enriched in both siderophile (e.g., Co, V, Mo, Mn) and chalcophile (e.g., Zn, Tl, As, Au, Pb, Cd) elements compared to cogenetic I-Py1 (Figs. 4, 5a). This elemental enrichment is characteristic of marcasite's lower formation temperature relative to pyrite, which results in the preferential incorporation of metals such as Zn, Tl, As, Pb, and Cd (Kim et al., 2006).

The Se/Tl ratios, which serve as indicators of fluid temperatures (Keith et al., 2016a, b), are low in both I-Mrc1 and I-Py1 (Fig. 8a), confirming that they were precipitated at low temperatures. However, their consistent values (medians of 0.29 and 0.35, respectively) suggest that factors beyond temperature gradients also significantly affect metal distribution. Rapid crystallization during fluid-seawater mixing elevates the specific surface-to-volume ratio of minerals, thereby enhancing their capacity for

metal incorporation through adsorption, inclusion, and isomorphism (Xu and Scott, 2005; Scott et al., 2009; Deditius et al., 2011). This process is evident in the elevated concentrations of elements like Co and As in I-Mrc1 (medians of 505 and 137 ppm, respectively), which exceed their concentrations in I-Py1 (medians of 71.9 and 79.6 ppm, respectively). The steep physicochemical gradients during the rapid cooling of fluids reduce metal solubility and drive the preferential deposition of metals into marcasite (Maslennikov et al., 2009; Keith et al., 2016a, b). Moreover, the orthorhombic crystal structure of marcasite, with its lattice deficiencies, may further facilitate the preferential incorporation of certain metals, such as As and Au, compared to pyrite characterized by a cubic lattice (Reich et al., 2005; Nestmeyer et al., 2021).

Rapid precipitation can lead to supersaturation of heavy metals such as Zn, Ag, and Pb, promoting the formation of mineral inclusions (Fig. 11a). The positive correlation between Zn and Cd ($r = 0.79$; Fig. 8c) suggests that Cd is primarily hosted in sphalerite inclusions, as indicated by the abrupt Zn spectra (Fig. 11a) (Cook et al., 2009). Similarly, Pb is likely associated with galena inclusions, which exhibit Tl enrichment (Fig. 8d) (Melekestseva et al., 2017). However, the subsequent low-temperature recrystallization from I-Mrc1 to I-Py1 releases these metals, forming abundant mineral inclusions, including galena, sphalerite, and Ag-rich phases in I-Py1 (Fig. 11b) (Wohlgemuth-Ueberwasser et al., 2015; Grant et al., 2018). This process may account for the Ag-Pb-rich layers within laminated I-Py1 (Fig. 6). Additionally, the preferential liberation of Pb relative to isomorphic Tl during recrystallization results in the decoupling of Pb and Tl at the boundary between I-Mrc1 and I-Py1 (Fig. 6).

During the precipitation of the intergrowths as hot fluids mix with seawater, Mn and Ba are incorporated as Mn-hydroxides and barium sulfate (BaSO_4) within I-Mrc1 (Fig. 8e), while Mo and V, derived from seawater (Fig. 8f), are preferentially adsorbed onto the surface of I-Mrc1 due to redox gradients (Maslennikov et al., 2017; Grant et al., 2018). This suggests that the partitioning of metals between pyrite and marcasite during rapid crystallization in a seawater environment is primarily governed by mineral crystallinity and physicochemical gradients (e.g., temperature and redox states).

5.2.2. Dissolution and reprecipitation of pyrite by marcasite

In the transition zone of the Xunmei chimneys, subhedral pyrite (I-Py2) was replaced by anhedral marcasite (I-Mrc2), forming assemblages (Type 2 Py-Mrc, Fig. 3c, d). The earlier pyrite (I-Py2) is characterized by significantly higher median concentrations of Au, Pb, As, Ag, Sb, and Zn, whereas the later marcasite (I-Mrc2) is enriched in Co, Mg, U, Tl, Ni, Ba, Cd, and Mn (Figs. 4, 5b).

The distinct co-enrichment of As, Sb, Zn, Au, Ag, and Pb in pyrite (I-Py2) may result from several processes such as magmatic degassing (Wohlgemuth-Ueberwasser et al., 2015), fluid boiling (Falkenberg et al., 2021), low-temperature precipitation (Maslennikov et al., 2009; Kristall et al., 2011), or evolution of fluid nature (Meng et al., 2020). However, the positive $\delta^{34}\text{S}$ values of I-Py2 (ranging from 1.93 to 2.96‰) and low $\text{Se/S} \times 10^6$ ratios ($1.56 - 22.0 \leq 500$) rule out magmatic degassing as a primary factor (Martin et al., 2020, 2022). I-Py2 is concentrated in both vapor-phase (e.g., As, Sb) and liquid-phase (e.g., Zn, Au, Ag, Pb) elements, without significant sulfur isotopic fractionation, which does not align with expectations for phase separation (Falkenberg et al., 2021; Nestmeyer et al., 2021). The low Se/Tl (median: 0.06 and 0.03) and Co/Ni (median: 6.12 and 9.20) ratios in both I-Py2 and I-Mrc2 (Fig. 8a, b) imply a decrease

in precipitation temperatures (Keith et al., 2016a, b; Meng et al., 2025), while the similarity in these ratios between I-Py2 and I-Mrc2 suggests that temperature gradients alone are insufficient to account for their chemical differences.

Marcasite (I-Mrc2), enriched in Mg, U, Ba, Mn, Tl, and Cd, is likely formed from low-temperature, oxidized fluids triggered by seawater infiltration inward chimneys. Specifically, Mg and U originate from seawater, Ba and Mn occur as barium sulfate and Mn-hydroxides, respectively, in a seawater environment, and Tl and Cd are rich in low-temperature fluids (Maslennikov et al., 2017; Meng et al., 2020). Oxidation has also been identified as a key factor for marcasite precipitation (Wu et al., 2018; Choi et al., 2021). In contrast, earlier pyrite (I-Py2) precipitated under relatively reducing conditions, resulting in the enrichment of hydrothermal metals, such as Au, Pb, As, Ag, Sb, and Zn (Choi et al., 2021; Wang et al., 2022a). Therefore, the evolution of fluid nature and composition is the most likely reason for metal partitioning between pyrite and marcasite. The high contents of Pb, Ag, and Zn in I-Py2 are likely associated with micro-inclusions such as galena, Ag-rich inclusions, and sphalerite, as indicated by the spikes in laser profiles (Fig. 11d). However, the smooth profiles of As, Au, and Sb suggest that these elements occur predominantly as solid solutions (Fig. 11d).

Elemental imaging of the reaction front between I-Py2 and I-Mrc2 reveals maxima in the concentrations of As, Ag, Au, Co, Pb, and Bi (Fig. 7), reflecting the accumulation of high trace element concentrations at local mineral-fluid interfacial layers (Putnis et al., 2005, 2009). Early-deposited pyrite (I-Py2) was replaced by later oxidized Fe-rich fluids (i.e., I-Mrc2). This replacement, involving dissolution and reprecipitation of I-Py2, created local non-equilibrium conditions that facilitated the remobilization and re-enrichment of metals (Ishida et al., 2022). This process induces metal deposition at the reaction front between I-Py2 and I-Mrc2 (Wang et al., 2022b) due to the steep physicochemical gradient (Putnis, 2009). This also leads to the formation of abundant mineral inclusions within I-Mrc2 (Fig. 11c). Similar reactions have been observed in the remobilization of Co and As within pyrites in the hydrothermal polymetallic deposit (Wang et al., 2022b). Spot analysis results, however, show that Co concentrations are higher in marcasite compared to pyrite. This may be attributed to the sharp temperature drop during marcasite precipitation, which favors the substitution of Co (and subsequently Ni) for Fe in the marcasite lattice (Maslennikov et al., 2009).

5.2.3. Replacement and recrystallization of marcasite by pyrite

At the Niaochao field, the transition layers of chimney fragments contain dense marcasite-pyrite pairs (Type 3 Py-Mrc, Fig. 3k-n) that formed through the replacement of earlier marcasite (II-Mrc2) by dense pyrite (II-Py2). II-Mrc2 is enriched in Mo, Mn, U, V, Mg, Cu, Cd, Tl, and Ni, whereas II-Py2 is concentrated in As, Pb, Ag, Au, and Sb (Figs. 4, 5c). The metal partitioning in this assemblage is comparable to that observed between I-Py2 and I-Mrc2 at the Xunmei field (Figs. 4, 5c).

The low Se/Tl and Co/Ni ratios in both II-Mrc2 and II-Py2 (Fig. 8a, b) are indicative of low-temperature precipitation conditions (Maslennikov et al., 2009; Meng et al., 2020). However, compared to II-Py2, earlier II-Mrc2 shows significantly higher concentrations of seawater-derived (e.g., Mo, Mg, V, U) and low-temperature-responsive (e.g., Mn, Cd, Tl, Ni) elements (Figs. 4, 5c) (Butler et al., 1999; Maslennikov et al., 2009; Choi et al., 2022). This suggests that II-Mrc2 precipitated under low-temperature and oxidizing conditions, possibly associated with

hydrothermal fluid-seawater mixing. The rapid interaction between hydrothermal fluids and seawater facilitates the incorporation of mineral inclusions, such as chalcopyrite and Mn-rich phases (Fig. 11e) (Maslennikov et al., 2009, 2017), leading to the enrichment of Cu and Mn in II-Mrc2. In contrast, chalcophile and precious metals like As, Pb, Sb, Ag, and Au, which preferentially precipitated in low-temperature and reducing conditions with limited seawater mixing (Keith et al., 2016a, b; Wang et al., 2022a), are significantly enriched in II-Py2 (Figs. 4, 5c). Therefore, the variations in trace elements between II-Mrc2 and II-Py2 are likely triggered by changes in fluid nature, analogous to the factor controlling metal distribution between I-Py2 and I-Mrc2 in the Xunmei vent field.

The marcasite-pyrite assemblages from the Niaochao field on the fast-spreading ridge generally exhibit lower concentrations of trace elements (except for Sb, Mn, and Pb) compared to those from the Xunmei field on the slow-spreading ridge (Figs. 4, 7) and published data of pyrites from Niaochao chimneys (Meng et al., 2025). Seafloor spreading rates can influence the degree and duration of fluid-rock interaction in the reaction zones of hydrothermal systems, leading to the formation of Ni-Mo-Se-rich SMS from fast-spreading ridges and Au-Ag-rich SMS from slow-spreading ridges (Knight et al., 2018; Toffolo et al., 2020). However, our findings contradict this model, indicating that the spreading rate is not the primary control on the chemical differences of marcasite-pyrite between the Niaochao and Xunmei vent fields. The marcasite-pyrite, characterized by a dense, coarse-grained texture (Fig. 3k-n), formed through low-temperature replacement processes (Knott et al., 1998; Grant et al., 2018). The relatively high contents of Sb, Mn, and Pb in this marcasite-pyrite pair confirm a low-temperature precipitation environment (Maslennikov et al., 2009). The replacement and recrystallization of II-Mrc2 by later II-Py2 may result in the liberation of trace elements, which are typically less soluble in pyrite as recrystallization proceeds (Wohlgemuth-Ueberwasser et al., 2015). When solubility limits are reached, these metals can precipitate as inclusions, such as galena, chalcopyrite, sphalerite, or Sb-bearing inclusions (e.g., tetrahedrite-tennantite) (Fig. 11f). Therefore, the recrystallization process likely accounts for the low metal content in the marcasite-pyrite from the Niaochao field.

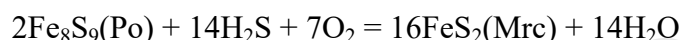
5.2.4. Seawater-induced alteration of pyrrhotite

Toward the middle zone of the Xunmei chimneys, columnar marcasite pseudomorphs (I-Mrc3) after pyrrhotite were precipitated alongside porous pyrites (I-Py3) (Type 4 Py-Mrc, Fig. 3e, f), a feature commonly observed in both fossil Urals sulfide deposits (Maslennikov et al., 2017) and modern vent fields (Cao et al., 2021; Choi et al., 2021). I-Mrc3 is characterized by elevated concentrations of Sb, Co, Zn, Cd, Au, Cu, and Ag, but lower abundances of Tl, V, Mn, Mo, Ba, and Mg compared to I-Py3 (Figs. 4, 5d).

The relatively higher Se/Tl (median: 2.13 and 17.0) and Co/Ni (median: 346 and 3794) ratios in both I-Py3 and I-Mrc3 compared to other pyrite-marcasite assemblages (Fig. 8a, b) suggest an increase in precipitation temperature and fluid activity as chimneys grow (Keith et al., 2016a, Meng et al., 2025). Nevertheless, minor seawater can still infiltrate into chimneys through porous walls, causing fluctuations in fluid compositions and physicochemical conditions (Tivey, 1995; Zhu et al., 2007). The high concentrations of seawater-derived elements, particularly vanadium, in the middle zone of the Xunmei chimney (Fig. 2a3) indicate that seawater mixes with hydrothermal

fluids within the chimney walls (Butler et al., 1999). This interaction creates low-temperature and oxidizing conditions, favoring the precipitation of porous I-Py3 with enrichment of seawater-derived (e.g., Mo, V, and Mg) and redox-sensitive (e.g., Mn and Ba) elements (Fig. 8e, f) (Maslennikov et al., 2009; Meng et al., 2020). The disequilibrium environment facilitates the formation of abundant mineral inclusions (Fig. 11h). Moreover, the reduction of seawater sulfate during fluid-seawater mixing elevates $\delta^{34}\text{S}$ values in I-Py3 (mean: 4.06‰, $n = 7$) (Fan et al., 2021), corresponding to high Mo and V contents (Fig. 13).

Marcasite (I-Mrc3) forms preferentially over pyrite when relatively low-temperature ($< 220^\circ\text{C}$) and oxidized fluids react with pyrrhotite (Po) (Knott et al., 1995; Choi et al., 2021). This process can be described by the following reaction equation, according to Qian et al. (2011):



The marcasite pseudomorphs (I-Mrc3) retain the columnar morphology of their pyrrhotite precursor (Fig. 3e, f). The alteration process triggered by fluid-seawater mixing induces the oversaturation of ion species in response to rapid physicochemical changes, thus driving the Au-Ag mineralization in the secondary marcasite (Choi et al., 2021). Furthermore, the enrichment of Au, Ag, Cu, Zn, Cd, and Sb during the transformation of pyrrhotite to marcasite can be attributed to adsorption processes, as evidenced by abundant mineral inclusions in I-Mrc3 (Fig. 11g). This finding aligns with the mechanism where trace elements can be adsorbed onto growing Fe-sulfides, a process controlled by the relative solubility of the metals, and are subsequently enriched in pyrite as it transforms from iron monosulfide (Gregory et al., 2014; 2015).

5.3. Sources and evolution of sulfur isotopes

The sulfur isotope data of sulfides from the Xunmei and Niaochao vent fields range from 0.72 to 6.41‰ (mean: 3.42‰, $n = 48$) and 2.32 to 7.57‰ (mean: 5.22‰, $n = 15$), respectively (Fig. 9a, b). These values fall within the typical $\delta^{34}\text{S}$ range (1 to 9‰, $n = 1841$) for mid-ocean ridge hydrothermal systems (Zeng et al., 2017). This suggests that sulfur is primarily derived from the leaching of igneous rocks ($\delta^{34}\text{S}$ value: 0.1 ± 0.5 ‰; Sakai et al., 1984) and the reduction of seawater sulfate ($\delta^{34}\text{S}$ value: ~ 20.9 ‰; Rees et al., 1978). The fractionation of sulfur isotopes during fluid-seawater mixing can be estimated using a two-endmember mixing model (One et al., 2007):

$$\delta^{34}\text{S}_{\text{mix}} = X \times \delta^{34}\text{S}_{\text{seawater}} + (1 - X) \times \delta^{34}\text{S}_{\text{basalt}}$$

where X is the percentage of seawater component, and $\delta^{34}\text{S}_{\text{mix}}$, $\delta^{34}\text{S}_{\text{seawater}}$ (~ 20.9 ‰), and $\delta^{34}\text{S}_{\text{basalt}}$ (~ 0.1 ‰) are the isotopic values of measured sulfides, seawater, and basalt, respectively. The sulfur isotopic compositions of sulfides at the Xunmei and Niaochao vent fields can be explained by the mixing of sulfur from seawater sulfate reduction (about 3% to 31% and 11% to 37%, respectively) and basaltic origin (about 69 to 97% and 63 to 89%, respectively). The detailed percentage of seawater-derived sulfur of sulfides within chimneys is shown in Fig. 12.

However, Janecky and Shanks (1988) noted that $\delta^{34}\text{S}$ values of fluids can only increase to a maximum of 4.5‰ when hydrothermal solutions mix with seawater in chimneys, owing to the limited reducing capacity to cause adequate sulfate reduction.

The $\delta^{34}\text{S}$ values of sulfides exceeding 4.5‰ at the Niaochao vent field suggest the involvement of additional processes, likely seafloor sulfate reduction driven by the consumption of Fe^{2+} released during basalt alteration in the feeder zone of hydrothermal systems (Petersen et al., 2000; Meng et al., 2022).

In the Xunmei chimneys, $\delta^{34}\text{S}$ values of pyrite and marcasite generally decrease from the interior to the exterior wall (from I-Py3 to I-Mrc1, Fig. 9c), which is contrary to the expected trend of increasing $\delta^{34}\text{S}$ values from inner to outer zones due to thermochemical sulfate reduction (TSR) (Fan et al., 2021; Meng et al., 2024). Molybdenum and V are indicative of seawater sources (Maslennikov et al., 2009), and their contents show a negative correlation with $\delta^{34}\text{S}$ values in pyrite and marcasite (excluding I-Py3) (Fig. 13). This suggests that sulfur isotope data cannot be solely explained by the simple mixing of sulfur from seawater sulfate and basaltic origin. The significant decrease in $\delta^{34}\text{S}$ values of colloform marcasite (as low as 0.72‰) in the outermost layer of the chimneys could be due to sulfur isotope exchange between fluid H_2S and seawater SO_4^{2-} during fluid-seawater mixing (Ono et al., 2007; Grosche et al., 2024). As chimneys grew inward, sulfur isotopic exchanges decreased, and TSR increased (Fig. 12), leading to slightly elevated $\delta^{34}\text{S}$ values in I-Py1 and I-Py2 (means: 3.04 and 2.41‰, respectively).

Moving inward, the anhedral marcasite (I-Mrc2) precipitated at low temperatures (Murowchick and Barnes, 1986) shows a wide range of $\delta^{34}\text{S}$ values (2.04 to 5.87‰). The low-temperature conditions limit thermochemical sulfate reduction (Ohmoto and Lasaga, 1982), but the dissolution and reduction of early-deposited anhydrite (Meng et al., 2021) could contribute isotopically heavy sulfur to the fluid (Janecky and Shanks, 1988; Ding et al., 2021). This mechanism increases the $\delta^{34}\text{S}$ values of I-Mrc2 (mean: 4.26‰, $n = 3$). In comparison, anhedral I-Py3 precipitated within chimney walls during discrete seawater infiltration (discussed in section 5.2.4), resulting in an elevation of $\delta^{34}\text{S}$ values via TSR (mean: 4.06‰, $n = 7$) and a concomitant enrichment of seawater-derived elements (e.g., Mo and V) (Fig. 13). The mixing of seawater and hot fluids facilitates the precipitation of anhydrite in a relative oxidizing condition. Seawater sulfate or anhydrite can be reduced through the oxidation of pyrrhotite during the inversion of pyrrhotite to marcasite (I-Mrc3), generating further high $\delta^{34}\text{S}$ values in the secondary marcasite, reaching up to 6.41‰ (Hu et al., 2022). The reduction of seawater sulfate or anhydrite could be the reason for the high $\delta^{34}\text{S}$ values of I-Ccp1 (mean: 3.86‰, $n = 3$) during hydrothermal overprinting.

As anhydrite was consumed and seawater was isolated, the $\delta^{34}\text{S}$ values of euhedral pyrite (I-Py4) and chalcopyrite (I-Ccp2) decreased (means: 3.52‰ and 3.09‰, respectively, Fig. 12) in the chimney conduits (Meng et al., 2022, 2024). Fan et al. (2021) reported negative $\delta^{34}\text{S}$ values for chalcopyrite (mean: -0.78‰, $n = 19$) at the Xunmei vent field, suggesting the possible contribution of magmatic volatiles. However, the positive $\delta^{34}\text{S}$ values of sulfides, coupled with low $\text{Se/S} \times 10^6$ ratios (almost all below 500), exclude the addition of magmatic sulfur in this study (Layton-Matthews et al., 2008; Martin et al., 2020).

In the Niaochao chimneys, the $\delta^{34}\text{S}$ values progressively increase from the exterior marcasite and pyrite to the interior chalcopyrite, with mean values rising from 4.54‰ and 4.69‰ to 6.92‰, respectively (Fig. 9c). The $\delta^{34}\text{S}$ value of chalcopyrite in the inner chimney likely reflects the fluid $\delta^{34}\text{S}$ value, given the minimal sulfur isotopic fractionation ($\Delta^{34}\text{S}_{\text{Cpy-H}_2\text{S}} = -0.1\text{‰}$) at $>300^\circ\text{C}$ (Ohmoto and Rye, 1979; McDermott et

al., 2015). However, the isotopic difference between marcasite-pyrite and chalcopyrite is as large as -5.52% (Table S4), which cannot be fully explained by equilibrium fractionation alone (Ohmoto and Rye, 1979; Syverson et al., 2015). The high $\delta^{34}\text{S}$ values in the fluids within the Niaochao field may be attributed to increases in sulfate reduction within the seafloor hydrothermal stockwork, as previously discussed. The progressive replacement of earlier sulfides by later hydrothermal fluids likely increases the proportion of later fluids, leading to gradually higher $\delta^{34}\text{S}$ values in the sulfides over time (Bluth and Ohmoto, 1988; Fan et al., 2021).

5.4. Factors and processes influencing marcasite formation

Marcasite is ubiquitous in global submarine vent fields, and its abundance varies significantly depending on host rocks, geological settings, and spreading rates (Table S5). Mafic-hosted vent fields typically exhibit higher concentrations of marcasite compared to ultramafic-hosted systems. Ultramafic-hosted vent fields, such as Kairei and Cheoem, are characterized by highly reducing conditions due to the production of H_2 and CH_4 during hydrothermal circulation (Wang et al., 2014; Choi et al., 2021). This environment favors the precipitation of minerals like chalcopyrite, isocubanite, and Fe-rich sphalerite, rather than marcasite, which requires relatively oxidizing conditions (Fouquet et al., 2010). Interestingly, the development of marcasite is not directly linked to the spreading rates of hydrothermal systems (Table S5), except for the ultraslow-spreading ridge, which typically hosts ultramafic-hosted vent fields with rare marcasite (Toffolo et al., 2020). In contrast, arc-related hydrothermal systems, which have lower temperatures, are more conducive to the formation of marcasite, often alongside minerals like sphalerite, galena, and barite (Fouquet et al., 1993; Kim et al., 2006; de Ronde et al., 2011).

Marcasite and its associated pyrite are commonly precipitated during low-temperature mineralization. Their mineralogical and geochemical characteristics provide a new window into the microchemical environment and metal remobilization in SMS deposits (Fig. 14). Details are outlined as follows:

i. Rapid crystallization during fluid-seawater mixing (Fig. 14b): Pyrite-marcasite, with colloform and anhedral textures, precipitates during rapid fluid-seawater mixing in the outer chimney walls, consistent with findings in the Xunmei (Dang et al., 2023), Mothra (Kim et al., 2006), TAG (Hannington et al., 1999), and Semenov-1 (Melekestseva et al., 2014) vent fields. This assemblage is enriched in seawater-derived elements like Mo, Mg, V, and U, as well as low-temperature-responsive elements such as Zn, Ag, Pb, and Tl. The precipitation of marcasite at low temperatures prevents thermochemical sulfate reduction (Zeng et al., 2017), while sulfur isotope exchanges during fluid-seawater mixing result in low $\delta^{34}\text{S}$ values in marcasite (Rouxel et al., 2008).

ii. Replacement and related metal redistribution (Fig. 14c): As chimneys grow, pyrite (marcasite) can be replaced by marcasite (pyrite) in SMS and VMS deposits, such as the EPR 11-13°N field (Graham et al., 1988), the TAG field (Knott et al., 1998), and the Urals sulfide deposit (Maslennikov et al., 2009). This replacement, occurring via dissolution and reprecipitation, leads to the redistribution of trace elements and the formation of metal-rich zones (e.g., As, Ag, Au, Pb). Additionally, the dissolution of anhydrite adds isotopically heavy sulfur to the fluids, resulting in marcasite with higher $\delta^{34}\text{S}$ values ($> 4.5\%$) (Ding et al., 2021).

iii. Pyrite-marcasite formation through recrystallization (Fig. 14d): In vent fields such as Niaochao and TAG (Grant et al., 2018), massive pyrites containing remnants of marcasite blades form via replacement and reprecipitation. This recrystallization process releases trace elements and leads to their low concentrations in marcasite-pyrite assemblages. Due to shorter upflow pathways, fluids in vents from fast-spreading ridges retain more heat than those from slow-spreading ridges (Syverson et al., 2023), potentially facilitating the replacement of early-deposited marcasite by later fluids. Although the recrystallization process can impact $\delta^{34}\text{S}$ variations in chimneys (Syverson et al., 2015), such phenomena are not observed in this study.

iv. Pyrrhotite transformation to marcasite (Fig. 14e): Pyrrhotite often transforms to marcasite instead of pyrite under low-temperature ($< 220^\circ\text{C}$) and oxidizing conditions (Qian et al., 2011), as seen in vent fields such as Galapagos Rift, Tianzuo, and Xunmei (Knott et al., 1995; Cao et al., 2021). The mineral transformation causes metal mineralization, including Sb, Co, Zn, Cd, Au, Cu, and Ag, via adsorption reaction (Gregory et al., 2014; 2015). Moreover, sulfate reduction via pyrrhotite oxidation can elevate $\delta^{34}\text{S}$ values in secondary marcasite (Hu et al., 2019).

v. Subseafloor fluid boiling and marcasite precipitation (Fig. 14f): According to Dang et al. (2023), coarse-grained marcasite, resulting from fluid boiling in the Xunmei vent field, displayed low Sb/Pb and Bi/Pb ratios but high As, Ag, Cu, and Pb contents. However, this specific type of marcasite was not found in this study. Fluid boiling in the subseafloor causes phase separation, resulting in the loss of aqueous H_2S and the generation of low-temperature, low-pH, vapor-rich fluids that promote marcasite precipitation (Nestmeyer et al., 2021). This process can produce significant sulfur isotopic fractionation in fluids (Falkenberg et al., 2021, 2022).

6. Conclusions

In-situ trace element and sulfur isotope analysis of pyrite-marcasite assemblages from the mafic-hosted Xunmei and Niaochao vent fields on the slow- and fast-spreading ridges, respectively, offers a deep comprehension of the metal partitioning and sulfur isotope variations between pyrite and marcasite during the low-temperature mineralization of SMS deposits.

Trace element characteristics reveal that recrystallization processes are the primary reason for the chemical differences within marcasite-pyrite between Niaochao and Xunmei vent fields. The detailed chemical characteristics of pyrite-marcasite pairs across chimneys are as follows: (i) Colloform marcasite-pyrite intergrowths arise from the rapid mixing of hot fluids and seawater, a process that creates higher surface-to-volume ratios in marcasite compared to pyrite, thereby favoring the enrichment of metals in marcasite. (ii) As marcasite replaces pyrite, processes involving dissolution and reprecipitation result in the remobilization of elements such as As, Ag, Au, Co, and Pb at their reaction front. (iii) Recrystallized marcasite-pyrite, formed through the replacement of marcasite by dense pyrite, is depleted in metals. (iv) The secondary marcasite, transformed from pyrrhotite under low-temperature and oxidizing conditions, is concentrated in metals such as Sb, Co, Zn, Cd, Au, Cu, and Ag. (v) Marcasite, induced by seafloor fluid boiling, is enriched in As, Ag, Cu, and Pb.

Sulfur isotope data indicate that sulfur in the Xunmei (mean: 3.42‰, $n = 48$) and Niaochao (mean: 5.22‰, $n = 15$) vent systems is distinctly affected by seawater sulfate reduction occurring in surface sulfide chimneys and subsurface hydrothermal stockworks, respectively. In the outermost chimney wall, sulfur isotope exchanges during fluid-seawater mixing result in decreased $\delta^{34}\text{S}$ values (as low as 0.72‰) of marcasite. Moving inward, the reduction of pre-existing anhydrite through the oxidation of fluid Fe^{2+} and pyrrhotite will increase $\delta^{34}\text{S}$ values (as high as 6.41‰) of marcasite. This process, coupled with the replacement of earlier sulfides by later high $\delta^{34}\text{S}$ (H_2S) fluids, progressively elevates $\delta^{34}\text{S}$ values across chimneys.

This study indicates that marcasite and its associated pyrite are more prevalent in mafic-hosted vent fields compared to reduced ultramafic-hosted systems. Their microchemical patterns emphasize that replacement and recrystallization play a crucial role in the redistribution of trace elements within submarine hydrothermal systems.

Data availability

Data are available through Mendeley Data at <https://doi.org/10.17632/2pgtjtvb4p.4>.

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Declaration of competing interests

The authors declare no competing interests.

Appendix A. Supplementary material

Supplementary research data: The data encompass detailed results of major elements (Table S1), trace elements of sulfides (Table S2) and reference standards (Table S3), and sulfur isotopes (Table S4).

Supplementary material: Figure S1 presents the major elements data; Figure S2 displays the EPMA elemental mapping of the marcasite-pyrite assemblage from the Niaochao vent field; Table S5 summarizes the formation of marcasite in global submarine hydrothermal systems.

References

- Alt, J.C., 1995. Subseafloor processes in mid-ocean ridge hydrothermal systems. *Geophysical Monograph American Geophysical Union, Washington, D.C.* pp. 85–114.
- Alt, J.C., Shanks III, W.C., Bach, W., Paulick, H., Garrido, C.J., Beaudoin, G., 2007. Hydrothermal alteration and microbial sulfate reduction in peridotite and gabbro exposed by detachment faulting at the Mid-Atlantic Ridge, 15°20'N (ODP Leg 209): A sulfur and oxygen isotope study. *Geochem. Geophys. Geosyst.* 8, 1–22.
- Bluth, G.J., Ohmoto, H., 1988. Sulfide-sulfate chimneys on the East Pacific Rise, 11° and 13°N latitudes. Part II: Sulfur isotopes. *Can. Mineral.* 26, 505–515.
- Burger, M., Gundlach-Graham, A., Allner, S., Schwarz, G., Wang, H.A.O., Gyr, L., Burgener, S., Hattendorf, B., Grolimund, D., Günther, D., 2015. High-Speed, High-Resolution, Multielemental LA-ICP-TOFMS Imaging: Part II. Critical Evaluation of Quantitative Three-Dimensional Imaging of Major, Minor, and Trace Elements in Geological Samples. *Anal. Chem.* 87, 8259–8267.
- Butler, I.B., Nesbitt, R.W., 1999. Trace element distributions in the chalcopyrite wall of a black smoker chimney: insights from laser ablation inductively coupled plasma mass spectrometry (LA-ICP-MS). *Earth Planet. Sci. Lett.* 167, 335–345.
- Cao, H., Sun, Z., Jiang, Z., Dong, A., Liu, W., 2021. Source origin and ore-controlling factors of hydrothermal sulfides from the Tianzuo hydrothermal field, Southwest Indian Ridge. *Ore Geol. Rev.* 134, 104168.
- Carbotte, S., Welch, S.M., Macdonald, K.C., 1991. Spreading rates, rift propagation, and fracture zone offset histories during the past 5 My on the Mid-Atlantic Ridge; 25°–27°30' S and 31°–34°30' S. *Mar. Geophys. Res.* 13, 51–80.
- Choi, S.K., Pak, S.J., Kim, J., Park, J.-W., Son, S.-K., 2021. Gold and tin mineralisation in the ultramafic-hosted Cheoeum vent field, Central Indian Ridge. *Mineral. Deposita* 56, 885–906.
- Choi, S.K., Pak, S.J., Park, J.-W., Kim, H.-S., Kim, J., Choi, S.-H., 2022. Trace-element distribution and ore-forming processes in Au–Ag-rich hydrothermal chimneys and mounds in the TA25 West vent field of the Tonga Arc. *Mineral. Deposita* 58, 135–160.
- Coltat, R., Boulvais, P., Riegler, T., Pelleret, E., Branqueta, Y., 2021. Element distribution in the root zone of ultramafic-hosted black smoker-like systems: Constraints from an Alpine analog. *Chem. Geol.* 559, 119916.
- Cook, N.J., Ciobanu, C.L., Pring, A., Skinner, W., Shimizu, M., Danyushevsky, L., Saini-Eidukat, B., Melcher, F., 2009. Trace and minor elements in sphalerite: A LA-ICPMS study. *Geochim. Cosmochim. Acta* 73, 4761–4791.
- Dang, Y., Li, C., Shi, X., Wang, S., Ye, J., Li, B., Yang, Y., Zhao, Q., Guan, Y., Mao, J., 2023. Metallogenetic process of Xunmei hydrothermal field (26°S), South Mid-Atlantic Ridge: Constraints from in-situ sulfur isotope and trace elements of sulfides. *Mar. Geol.* 466, 107182.
- de Ronde, C., Massoth, G.J., Butterfield, D.A., Christenson, B.W., Ishibashi, J., Ditchburn, R.G., Hannington, M., Brathwaite, R.L., Lupton, J.E., Kamenetsky, V.S., 2011. Submarine hydrothermal activity and gold-rich mineralization at Brothers Volcano, Kermadec Arc, New Zealand. *Mineral. Deposita* 46, 541–584.
- Deditius, A.P., Utsunomiya, S., Reich, M., Kesler, S.E., Ewing, R.C., Hough, R., Walshe, J., 2011. Trace metal nanoparticles in pyrite. *Ore Geol. Rev.* 42, 32–46.

- 771 Ding, T., Tao, C., Dias, Á.A., Liang, J., Chen, J., Wu, B., Ma, D., Zhang, R., Wang, J., Liao, S.,
772 Wang, Y., Yang, W., Liu, J., Li, W., Zhang, G., Huang, H., 2021. Sulfur isotopic compositions
773 of sulfides along the Southwest Indian Ridge: Implications for mineralization in ultramafic
774 rocks. *Mineral. Deposita* 56, 991–1006.
- 775 Falkenberg, J.J., Keith, M., Haase, K.M., Bach, W., Klemm, R., Strauss, H., Yeo, I.A., Rubin, K.H.,
776 Storch, B., Anderson, M.O., 2021. Effects of fluid boiling on Au and volatile element
777 enrichment in submarine arc-related hydrothermal systems. *Geochim. Cosmochim. Acta* 307,
778 105–132.
- 779 Falkenberg, J.J., Keith, M., Haase, K.M., Sporer, C., Bach, W., Klemm, R., Strauss, H., Storch, B.,
780 Peters, C., Rubin, K.H., Anderson, M.O., 2022. Spatial variations in magmatic volatile influx
781 and fluid boiling in submarine hydrothermal systems of Niutahi caldera, Tonga rear-arc.
782 *Geochem. Geophys. Geosyst.* 23, e2021GC010259.
- 783 Fan, L., Wang, G., Astrid, H., Basem, Z., Shi, X., 2021. Sulfur and copper isotopic composition of
784 seafloor massive sulfides and fluid evolution in the 26°S hydrothermal field, Southern Mid-
785 Atlantic Ridge. *Mar. Geol.* 435, 106436.
- 786 Fan, L., Wang, G., Astrid, H., Basem, Z., Shi, X., Lei, Q., 2022. Systematic variations in trace
787 element composition of pyrites from the 26° S hydrothermal field, Mid-Atlantic Ridge. *Ore*
788 *Geol. Rev.* 148, 105006.
- 789 Flude, S., Haschke, M., Storey, M., 2017. Application of benchtop micro-XRF to geological
790 materials. *Min. Mag.* 81, 923–948.
- 791 Fouquet, Y., Cambon, P., Etoubleau, J., Charlou, J.L., Ondréas, H., Barriga, F.J.A.S., Cherkashov,
792 G., Semkova, T., Poroshina, I., Bohn, M., 2010. Geodiversity of hydrothermal processes along
793 the Mid-Atlantic Ridge and ultramafic-hosted mineralization: A new type of oceanic Cu-Zn-
794 Co-Au volcanogenic massive sulfide deposit. *Geophys. Monogr. Ser.* 188, 321–367.
- 795 Fu, J., Hu, Z., Zhang, W., Yang, L., Liu, Y., Li, M., Zong, K., Gao, S., Hu, S., 2016. In situ sulfur
796 isotopes ($\delta^{34}\text{S}$ and $\delta^{33}\text{S}$) analyses in sulfides and elemental sulfur using high sensitivity cones
797 combined with the addition of nitrogen by laser ablation MC-ICP-MS. *Anal. Chim. Acta* 911,
798 14–26.
- 799 Gao, J.F., Jackson, S.E., Dubé, B., Kontak, D.J., De Souza, S., 2015. Genesis of the Canadian
800 Malartic, Côte Gold, and Musselwhite gold deposits: insights from LA-ICP-MS element
801 mapping of pyrite. Open File 7852. In: Dubé, B., Mercier-Langevin, P. (Eds.), Targeted
802 Geoscience Initiative 4: Contributions to the Understanding of Precambrian Lode Gold
803 Deposits and Implications for Exploration. Geological Survey of Canada, pp. 157–175.
- 804 George, L.L., Cook, N.J., Crowe, B.B.P., Ciobanu, C.L., 2018. Trace elements in hydrothermal
805 chalcopyrite. *Min. Mag.* 82, 59–88.
- 806 Graham, U.M., Bluth, G.J., Ohmoto, H., 1988. Sulfide-sulfate chimneys on the East Pacific Rise,
807 11° and 13°N latitudes; Part I, Mineralogy and paragenesis. *Can. Mineral.* 26, 487–504.
- 808 Grant, H., Hannington, M.D., Petersen, S., Frische, M., Fuchs, S., 2018. Constraints on the behavior
809 of trace elements in the actively-forming TAG deposit, Mid-Atlantic Ridge, based on LA-ICP-
810 MS analyses of pyrite. *Chem. Geol.* 498, 45–71.
- 811 Gregory, D., Large, R.R., Halpin, J.A., Baturina, E.L., Lyons, T.W., Wu, S., Danyushevsky, L.V.,
812 Sack, P.J., Chappaz, A., Maslennikov, V.V., 2015. Trace Element Content of Sedimentary
813 Pyrite in Black Shales. *Econ. Geol.* 110, 1389–1410.
- 814 Gregory, D., Meffre, S., Large, R., 2014. Comparison of metal enrichment in pyrite framboids from

- 815 a metal-enriched and metal-poor estuary. *Am. Mineral.* 99, 633–644.
- 816 Grosche, A., Keith, M., Klemm, R., Strauss, H., Rosca, C., König, S., 2024. Temperature-controlled
817 Se-S isotope fractionation during seawater mixing and sulfide precipitation in black smoker
818 chimneys. *Geochim. Cosmochim. Acta* 372, 13–27.
- 819 Hannington, M.D., Bleeker, W., Kjarsgaard, B., 1999. Sulfide mineralogy, geochemistry, and ore
820 genesis of the Kidd Creek deposit: Part II. In: Hannington, M.D., Barrie, C.T. (Eds.), *The Giant
821 Kidd Creek Volcanogenic Massive Sulfide Deposit, Western Abitibi Subprovince*. Society of
822 Economic Geologists, Canada, pp. 225–266.
- 823 Hannington, M.D., Jonasson, I.R., Herzig, P.M., Petersen, S., 1995. Physical and Chemical
824 Processes of Seafloor Mineralization at Mid-Ocean Ridges. Washington DC American
825 Geophysical Union Geophysical Monograph Series 91, 115–157.
- 826 Haymon, R.M., 1983. Growth history of hydrothermal black smoker chimneys. *Nature* 301, 695–
827 698.
- 828 Herzig, P.M., Hannington, M.D., Arribas Jr, A., 1998. Sulfur isotopic composition of hydrothermal
829 precipitates from the Lau back-arc: implications for magmatic contributions to seafloor
830 hydrothermal systems. *Mineral. Deposita* 33, 226–237.
- 831 Holler, G., Marchig, V., 1990. Hydrothermal activity on the East Pacific Rise: Stages of
832 development. *Geologisch. Jahrbuch Reihe B75*, 3–22.
- 833 Hu, S.-Y., Wang, X.-C., Tian, L., Martin, L., Schoneveld, L., Barnes, S.J., Guagliardo, P., Ding,
834 W., Rickard, W.D., 2022. Variability of sulfur isotopes and trace metals in pyrites from the
835 upper oceanic crust of the South China Sea basin, implications for sulfur and trace metal
836 cycling in subsurface. *Chem. Geol.* 606, 120982.
- 837 Ishida, M., Romero, R., Leisen, M., Yasukawa, K., Kato, Y., 2022. Auriferous pyrite formed by
838 episodic fluid inputs in the Akeshi and Kasuga high-sulfidation deposits, Southern Kyushu,
839 Japan. *Mineral. Deposita* 57, 129–145.
- 840 Janecky, D.R., Shanks III, W.C., 1988. Computational modeling of chemical and sulfur isotopic
841 reaction processes in seafloor hydrothermal systems: chimneys, massive sulfides, and
842 subjacent alteration zones. *Can. Mineral.* 26, 805–825.
- 843 Kajiwar, Y., Krouse, H.R., 1971. Sulfur Isotope Partitioning in Metallic Sulfide Systems. *Canadian
844 Geophys. Monogr. Ser.* 8, 1397–1408.
- 845 Keith, M., Haase, K.M., Klemm, R., Krumm, S., Strauss, H., 2016a. Systematic variations in the
846 trace element and sulfur isotope composition of pyrite with stratigraphic depth in the
847 Skouriotissa volcanic-hosted massive sulfide deposit, Troodos ophiolite, Cyprus. *Chem. Geol.*
848 423, 7–18.
- 849 Keith, M., Haase, K.M., Schwarzschaemper, U., Klemm, R., Petersen, S., Bach, W., 2014. Effects
850 of temperature, sulfur, and oxygen fugacity on the composition of sphalerite from submarine
851 hydrothermal vents. *Geology* 42, 699–702.
- 852 Keith, M., Häckel, F., Haase, K.M., Schwarz-Schamper, U., Klemm, R., 2016b. Trace element
853 systematics of pyrite from submarine hydrothermal vents. *Ore Geol. Rev.* 72, 728–745.
- 854 Kim, J., Lee, I., Halbach, P., Lee, K.-Y., Ko, Y.-T., Kim, K.-H., 2006. Formation of hydrothermal
855 vents in the North Fiji Basin: Sulfur and lead isotope constraints. *Chem. Geol.* 233, 257–275.
- 856 Knight, R.D., Roberts, S., Webber, A.P., 2018. The influence of spreading rate, basement

- composition, fluid chemistry, and chimney morphology on the formation of gold-rich SMS deposits at slow and ultraslow mid-ocean ridges. *Mineral. Deposita* 53, 143–152.
- Knott, R., Fallick, A.E., Rickard, D., Bäcker, H., 1995. Mineralogy and sulphur isotope characteristics of a massive sulphide boulder, Galapagos Rift, 85° 55' W. Geological Society, London, Special Publications 87, 207–222.
- Knott, R., Fouquet, Y., Honnorez, J., Petersen, S., Bohn, M., 1998. Petrology of hydrothermal mineralization: a vertical section through the TAG mound. *Proc. ODP Sci. Results* 158, 5–26.
- Kristall, B., Nielsen, D., Hannington, M.D., Kelley, D.S., Delaney, J.R., 2011. Chemical microenvironments within sulfide structures from the Mothra Hydrothermal Field: Evidence from high-resolution zoning of trace elements. *Chem. Geol.* 290, 12–30.
- Layton-Matthews, D., Peter, J.M., Scott, S.D., Leybourne, M.I., 2008. Distribution, Mineralogy, and Geochemistry of Selenium in Felsic Volcanic-Hosted Massive Sulfide Deposits of the Finlayson Lake District, Yukon Territory, Canada. *Econ. Geol.* 103, 61–88.
- Lehrmann, B., Cooper, M.J., Milton, J.A., Murton, B.J., 2022. Formation, remobilisation and alteration processes at inactive hydrothermal vents: insights from elemental analysis of Cu-(Fe-) S sulfides from TAG, Mid-Atlantic Ridge. *Mineral. Deposita* 57, 1431–1448.
- Li, T., Yang, Y., Wang, G., Fan, L., Wang, C., Bing, L., 2014. The Mineralogical Characteristics of Pyrite at 26°S Hydrothermal Field, South Mid-Atlantic Ridge. *Acta Geol. Sin.* 88, 179–180.
- Liu, Y., Hu, Z., Gao, S., Günther, D., Xu, J., 2008. In situ analysis of major and trace elements of anhydrous minerals by LA-ICP-MS without applying an internal standard. *Chem. Geol.* 257, 34–43.
- Macdonald, K.C., Fox, P.J., 1983. Overlapping spreading centres: new accretion geometry on the East Pacific Rise. *Nature* 302, 55–58.
- Mao, Y., Qin, K., Barnes, S.J., Ferraina, C., Iacono-Marziano, G., Verrall, M., Tang, D., Xue, S., 2017. A revised oxygen barometry in sulfide-saturated magmas and application to the Permian magmatic Ni–Cu deposits in the southern Central Asian Orogenic Belt. *Mineral. Deposita* 53, 731–755.
- Martin, A.J., Keith, M., Parvaz, D.B., McDonald, I., Boyce, A.J., Mcfall, K., Jenkin, G.R.T., Strauss, H., Macleod, C.J., 2020. Effects of magmatic volatile influx in mafic VMS hydrothermal systems: Evidence from the Troodos ophiolite, Cyprus. *Chem. Geol.* 531, 119325.
- Martin, A.J., McDonald, I., Jamieson, J.W., Jenkin, G.R., McFall, K.A., Piercey, G., MacLeod, C.J., Layne, G.D., 2022. Mineral-scale variation in the trace metal and sulfur isotope composition of pyrite: implications for metal and sulfur sources in mafic VMS deposits. *Mineral. Deposita* 57, 911–933.
- Maslennikov, V.V., Maslennikova, S.P., Large, R.R., Danyushevsky, L.V., 2009. Study of Trace Element Zonation in Vent Chimneys from the Silurian Yaman-Kasy Volcanic-Hosted Massive Sulfide Deposit (Southern Urals, Russia) Using Laser Ablation-Inductively Coupled Plasma Mass Spectrometry (LA-ICPMS). *Econ. Geol.* 104, 1111–1141.
- Maslennikov, V.V., Maslennikova, S.P., Large, R.R., Danyushevsky, L.V., Herrington, R.J., Ayupova, N.R., Zaykov, V.V., Lein, A.Y., Tseluyko, A.S., Melekestseva, I.Y., 2017. Chimneys in Paleozoic massive sulfide mounds of the Urals VMS deposits: Mineral and trace element comparison with modern black, grey, white and clear smokers. *Ore Geol. Rev.* 85, 64–106.

- 900 McDermott, J.M., Ono, S., Tivey, M.K., Seewald, J.S., Shanks III, W.C., Solow, A.R., 2015.
 901 Identification of sulfur sources and isotopic equilibria in submarine hot-springs using multiple
 902 sulfur isotopes. *Geochim. Cosmochim. Acta* 160, 169–187.
- 903 Melekestseva, I.Y., Maslennikov, V.V., Tretyakov, G.A., Nimis, P., Beltenev, V.E.,
 904 Rozhdestvenskaya, I., Maslennikova, S.P., Belogub, E.V., Danyushevsky, L.V., Large, R.R.,
 905 2017. Gold- and silver-rich massive sulfides from the Semenov-2 hydrothermal field,
 906 13°31.13'N, Mid-Atlantic Ridge: A case of magmatic contribution? *Econ. Geol.* 112, 741–773.
- 907 Melekestseva, I.Y., Tretyakov, G.A., Nimis, P., Yuminov, A.M., Maslennikov, V.V.,
 908 Maslennikova, S.P., Kotlyarov, V.A., Beltenev, V.E., Danyushevsky, L.V., Large, R., 2014.
 909 Barite-rich massive sulfides from the Semenov-1 hydrothermal field (Mid-Atlantic Ridge,
 910 13°30.87' N): Evidence for phase separation and magmatic input. *Mar. Geol.* 349, 37–54.
- 911 Meng, X., Jin, X., Li, X., Chu, F., Zhang, W., Wang, H., Zhu, J., Li, Z., 2021. Mineralogy and
 912 geochemistry of secondary minerals and oxyhydroxides from the Xunmei hydrothermal field,
 913 Southern Mid-Atlantic Ridge (26°S): Insights for metal mobilization during the oxidation of
 914 submarine sulfides. *Mar. Geol.* 442, 106654.
- 915 Meng, X., Jin, X., Li, X., Chu, F., Zhu, J., Wang, Y., Zhou, P., 2022. Subseafloor mineralization
 916 related to the shallow seawater-hydrothermal circulation system in the Longqi hydrothermal
 917 field, Southwest Indian Ridge (49.6°E): Evidence from in situ trace element and sulfur isotope
 918 compositions of pyrite varieties. *Ore Geol. Rev.* 145, 104914.
- 919 Meng, X., Li, X., Chu, F., Zhu, J., Lei, J., Li, Z., Wang, H., Chen, L., Zhu, Z., 2020. Trace element
 920 and sulfur isotope compositions for pyrite across the mineralization zones of a sulfide chimney
 921 from the East Pacific Rise (1-2°S). *Ore Geol. Rev.* 116, 103209.
- 922 Meng, X., Li, X., Holzheid, A., Zoheir, B., Chen, L., Frische, M., Rout, S.S., Li, W., Chu, F., Jin,
 923 X., 2025. From formation to maturation: Trace element systematics of sulfide chimneys from
 924 the Niochao vent field (East Pacific Rise). *Chem. Geol.* 674, 122584.
- 925 Meng, X., Li, X., Holzheid, A., Zoheir, B., Jin, X., Li, Z., Wang, H., Zhu, Z., Li, J., Zhao, J., Chu,
 926 F., 2024. Sulfur isotopes as a new indicator for the maturation of sulfide chimneys in submarine
 927 hydrothermal systems. *Mar. Geol.* 467, 107191.
- 928 Murowchick, J.B., 1992. Marcasite inversion and the petrographic determination of pyrite ancestry.
 929 *Econ. Geol.* 87, 1141–1152.
- 930 Murowchick, J.B., Barnes, H.L., 1986. Marcasite precipitation from hydrothermal solutions.
 931 *Geochim. Cosmochim. Acta* 50, 2615–2629.
- 932 Nestmeyer, M., Keith, M., Haase, K.M., Klemm, R., Voudouris, P., Schwarz-Schampera, U., Strauss,
 933 H., Kati, M., Magganis, A., 2021. Trace element signatures in pyrite and marcasite from
 934 shallow marine island arc-related hydrothermal vents, Calypso Vents, New Zealand, and
 935 Paleochori Bay, Greece. *Front. Earth Sci.* 9, 641654.
- 936 Niu, Y., Rodey, B., 1994. Magmatic processes at a slow spreading ridge segment: 26°S Mid-Atlantic
 937 Ridge. *J. Geophys. Res. Solid Earth* 99, 19719–19740.
- 938 Ohmoto, H., Rye, R.O., 1979. Isotopes of sulfur and carbon. In: Barnes, H.L. (Ed.), *Geochemistry
 939 of Hydrothermal Ore Deposits*. John Wiley and Sons Ltd, New York, pp. 506–567.
- 940 Ono, S., Shanks III, W.C., Rouxel, O.J., Rumble, D., 2007. S-33 constraints on the seawater sulfate
 941 contribution in modern seafloor hydrothermal vent sulfides. *Geochim. Cosmochim. Acta* 71,
 942 1170–1182.

- 943 Petersen, S., Herzig, P.M., Hannington, M.D., 2000. Third dimension of a presently forming VMS
944 deposit: TAG hydrothermal mound, Mid-Atlantic Ridge, 26°N. *Mineral. Deposita* 35, 233–
945 259.
- 946 Petersen, S., Lehrmann, B., Murton, B.J., 2018. Modern seafloor hydrothermal systems: New
947 perspectives on ancient ore-forming processes. *Elements* 14, 307–312.
- 948 Putnis, A., 2002. Mineral replacement reactions: from macroscopic observations to microscopic
949 mechanisms. *Mineral. Mag.* 66, 689–708.
- 950 Putnis, A., 2009. Mineral replacement reactions. *Rev. Mineral. Geochem.* 70, 87–124.
- 951 Qian, G., Fang, X., Brugger, J., Skinner, W.M., Pring, A., 2011. Replacement of pyrrhotite by pyrite
952 and marcasite under hydrothermal conditions up to 220 °C: An experimental study of reaction
953 textures and mechanisms. *Am. Mineral.* 96, 1878–1893.
- 954 Rees, C.E., Jenkins, W.J., Monster, J., 1978. The sulphur isotopic composition of ocean water
955 sulphate. *Geochim. Cosmochim. Acta* 42, 377–381.
- 956 Reich, M., Kesler, S.E., Utsunomiya, S., Palenik, C.S., Chrysosoulis, S.L., Ewing, R.C., 2005.
957 Solubility of gold in arsenian pyrite. *Geochim. Cosmochim. Acta* 69, 2781–2796.
- 958 Román, N., Reich, M., Leisen, M., Morata, D., Barra, F., Deditius, A.P., 2019. Geochemical and
959 micro-textural fingerprints of boiling in pyrite. *Geochim. Cosmochim. Acta* 246, 60–85.
- 960 Rouxel, O., Fouquet, Y., Ludden, J.N., 2004. Subsurface processes at the lucky strike hydrothermal
961 field, Mid-Atlantic ridge: evidence from sulfur, selenium, and iron isotopes. *Geochim.*
962 *Cosmochim. Acta* 68, 2295–2311.
- 963 Rouxel, O., Shanks III, W.C.S., Bach, W., Edwards, K.J., 2008. Integrated Fe- and S-isotope study
964 of seafloor hydrothermal vents at East Pacific Rise 9–10°N. *Chem. Geol.* 252, 214–227.
- 965 Sakai, H., Marais, D.J.D., Ueda, A., Moore, J.G., 1984. Concentrations and isotope ratios of carbon,
966 nitrogen and sulfur in ocean-floor basalts. *Geochim. Cosmochim. Acta* 48, 2433–2441.
- 967 Scott, R.J., Meffre, S., Woodhead, J., Gilbert, S.E., Berry, R.F., Emsbo, P., 2009. Development of
968 framboidal pyrite during diagenesis, low-grade regional metamorphism, and hydrothermal
969 alteration. *Econ. Geol.* 104, 1143–1168.
- 970 Syverson, D.D., Awolayo, A.N., Tutolo, B.M., 2023. Seafloor spreading and the delivery of sulfur
971 and metals to Earth's oceans. *Geology* 51, 1168–1172.
- 972 Syverson, D.D., Ono, S., Shanks III, W.C., Seyfried Jr., W.E.S., 2015. Multiple sulfur isotope
973 fractionation and mass transfer processes during pyrite precipitation and recrystallization: An
974 experimental study at 300 and 350 °C. *Geochim. Cosmochim. Acta* 165, 418–434.
- 975 Tao, C., Lin, J., Wu, G., German, C.R., Yoerger, D.R., Chen, Y.J., Guo, S., Zeng, Z., Han, X., Zhou,
976 N., 2008. First Active Hydrothermal Vent Fields Discovered at the Equatorial Southern East
977 Pacific Rise. *Am. Geophys. Union Fall Meet. 2008 Abstract #V41B-V2081*.
- 978 Tivey, M.K., 1995. Modeling Chimney Growth and Associated Fluid Flow at Seafloor
979 Hydrothermal Vent Sites. *Geophys. Monogr. Ser.* 91, 158–177.
- 980 Tivey, M.K., 1998. How to build a black smoker chimney. *Oceanus* 41, 22–26.
- 981 Toffolo, L., Nimis, P., Tretyakov, G.A., Melekestseva, I.Y., Beltenev, V.E., 2020. Seafloor massive
982 sulfides from mid-ocean ridges: Exploring the causes of their geochemical variability with

- 983 multivariate analysis. *Earth-Sci. Rev.* 201, 102958.
- 984 Wang, S., Li, C., Li, B., Dang, Y., Shi, X., 2022a. Constraints on fluid evolution and growth
985 processes of black smoker chimneys by pyrite geochemistry: A case study of the Tongguan
986 hydrothermal field, South Mid-Atlantic Ridge. *Ore Geol. Rev.* 140, 104410.
- 987 Wang, S., Sun, W., Huang, J., Zhai, S., Li, H., 2020. Coupled Fe-S isotope composition of sulfide
988 chimneys dominated by temperature heterogeneity in seafloor hydrothermal systems. *Sci. Bull.*
989 65, 1767–1774.
- 990 Wang, Y., Han, X., Petersen, S., Jin, X., Qiu, Z., Zhu, J., 2014. Mineralogy and geochemistry of
991 hydrothermal precipitates from Kaiei hydrothermal field, Central Indian Ridge. *Mar. Geol.*
992 354, 69–80.
- 993 Wang, Z., Wang, Y., Peng, E., Zou, S., Deng, T., Lai, F., Ning, J., Dong, G., Xu, D., 2022b. Micro-
994 textural and chemical fingerprints of hydrothermal cobalt enrichment in the Jingchong Co-Cu
995 polymetallic deposit, South China. *Ore Geol. Rev.* 142, 104721.
- 996 Wohlgemuth-Ueberwasser, C.C., Viljoen, F., Petersen, S., Vorster, C., 2015. Distribution and
997 solubility limits of trace elements in hydrothermal black smoker sulfides: An in-situ LA-ICP-
998 MS study. *Geochim. Cosmochim. Acta* 159, 16–41.
- 999 Wu, Y., Evans, K., Li, J., Fougereuse, D., Large, R.R., Guagliardo, P., 2019. Metal remobilization
1000 and ore-fluid perturbation during episodic replacement of auriferous pyrite from an epizonal
1001 orogenic gold deposit. *Geochim. Cosmochim. Acta* 245, 98–117.
- 1002 Wu, Y., Li, J., Evans, K., Koenig, A.E., Li, Z.-K., O'Brien, H., Lahaye, Y., Rempel, K., Hu, S.,
1003 Zhang, Z., Jun, P., 2018. Ore-Forming Processes of the Daqiao Epizonal Orogenic Gold
1004 Deposit, West Qinling Orogen, China: Constraints from Textures, Trace Elements, and Sulfur
1005 Isotopes of Pyrite and Marcasite, and Raman Spectroscopy of Carbonaceous Material. *Econ.*
1006 *Geol.* 113, 1093–1132.
- 1007 Xu, Q., Scott, S.D., 2005. Spherulitic pyrite in seafloor hydrothermal deposits: Products of rapid
1008 crystallization from mixing fluids. *Biennial SGA Meeting 6-29*, 711–713.
- 1009 Yao, X., Xia, F., Deditius, A., Brugger, J., Etschmann, B., Pearce, M.A., Pring, A., 2020. The
1010 mechanism and kinetics of the transformation from marcasite to pyrite: in situ and ex situ
1011 experiments and geological implications. *Contrib. Mineral. Petrol.* 175, 27.
- 1012 Zeng, Z., Ma, Y., Chen, S., Selby, D., Wang, X., Yin, X., 2017. Sulfur and lead isotopic
1013 compositions of massive sulfides from deep-sea hydrothermal systems: Implications for ore
1014 genesis and fluid circulation. *Ore Geol. Rev.* 87, 155–171.
- 1015 Zeng, Z., Ma, Y., Yin, X., Selby, D., Kong, F., Chen, S., 2015a. Factors affecting the rare earth
1016 element compositions in massive sulfides from deep-sea hydrothermal systems. *Geochem.*
1017 *Geophys. Geosyst.* 16, 2679–2693.
- 1018 Zeng, Z., Niedermann, S., Chen, S., Wang, X., Li, Z., 2015b. Noble gases in sulfide deposits of
1019 modern deep-sea hydrothermal systems: Implications for heat fluxes and hydrothermal fluid
1020 processes. *Chem. Geol.* 409, 1–11.
- 1021 Zhang, Y., Chu, F., Li, Z., Dong, Y., Wang, H., Li, X., Long, J., 2020. Gold enrichment in
1022 hydrothermal sulfides from the Okinawa Trough: An in-situ LA-ICP-MS study. *Ore Geol. Rev.*
1023 116, 103255.
- 1024 Zhu, W., Tivey, M.K., Gittings, H., Craddock, P.R., 2007. Permeability-porosity relationships in
1025 seafloor vent deposits: Dependence on pore evolution processes. *J. Geophys. Res. Solid Earth*

1026 112, 2637-2655.

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Figure captions

Fig. 1. (a) Location of the Xunmei and Niaochao hydrothermal fields (GeoMapApp, www.geomapapp.org), (b) Map of the Xunmei vent field on the Southern Mid-Atlantic Ridge (SMAR) and the collected samples, (c) Map of the Niaochao vent field on the East Pacific Rise (EPR) and the collected samples.

Fig. 2. Trace element mapping of transverse sections from representative sulfide chimneys and sulfide fragments from the Niaochao (a1-3) and Xunmei (b1-3) vent fields by the micro-XRF.

Fig. 3. Petrographic features of sulfides from the Xunmei (a-i) and Niaochao (j-o) vent fields. (a) Colloform marcasite (I-Mrc1) and anhedral pyrite (I-Py1) occur as cogenetic intergrowths in the outermost zone. (b) Back-scattered electron (BSE) image shows marcasite-pyrite intergrowths. (c) Subhedral pyrite (I-Py2) was surrounded by anhedral marcasite (I-Mrc2) in the transition zone. (d) BSE image of pyrite and marcasite aggregates. (e) Columnar marcasite pseudomorphs (I-Mrc3) after pyrrhotite were enveloped by porous anhedral pyrite (I-Py3) in the middle zone. (f) BSE image of marcasite pseudomorphs altered by Fe-Si oxyhydroxides (Fe-Si oxy). (g) Sphalerite (I-Sp) was replaced by pyrite and chalcopyrite (I-Ccp1). (h) Anhedral pyrite (I-Py3) was overgrown by euhedral pyrite (I-Py4). (i) Granular pyrite (I-Py4) was surrounded by massive chalcopyrite (I-Ccp2) in the interior zone. (j) Fine-grained pyrite (II-Py1) was rimmed by colloform marcasite (II-Mrc1) in the outermost layer. (k) Anhedral marcasite (II-Mrc2) was replaced by dense pyrite (II-Py2) in the transition layer. (l) Pyrite-marcasite assemblages were formed from the replacement of marcasite by pyrite. (m) Marcasite blades within massive pyrite grew into veined forms. (n) BSE image of marcasite and pyrite aggregates. (o) Massive chalcopyrite (II-Ccp) replaced earlier pyrite in the interior layer.

Fig. 4. Comparative box plots of representative trace elements in pyrite (Py) and marcasite (Mrc) from the Xunmei (yellow) and Niaochao (red) vent fields. The plots illustrate the distribution, median values, and variability of key trace elements (ppm).

Fig. 5. Spider diagrams illustrate the comparison of trace element contents between paired marcasite (Mrc) and pyrite (Py) from the Xunmei (a, b, d) and Niaochao (c) vent fields, including (a) I-Mrc1 vs. I-Py1, (b) I-Mrc2 vs. I-Py2, (c) II-Mrc2 vs. II-Py2, and (d) I-Mrc3 vs. I-Py3.

Fig. 6. A reflected light photomicrograph and images of LA-ICP-TOF-MS elemental mapping (in counts per second) of the marcasite-pyrite intergrowth (I-Mrc1 and I-Py1) from the Xunmei chimneys.

Fig. 7. Reflected light and BSE photomicrographs and elemental mapping of pyrite (I-Py2) and marcasite (I-Mrc2) pairs from the Xunmei chimneys. Arsenic is determined via EPMA and LA-ICP-TOF-MS, and other metals are detected using LA-ICP-TOF-MS (in counts per second).

Fig. 8. Concentrations of (a) Tl vs. Se, (b) Ni vs. Co, (c) Cd vs. Zn, (d) Pb vs. Tl, (e) Mg vs. Mn, and (f) Mo vs. V in pyrite (Py) and marcasite (Mrc) from the Xunmei and Niaochao vent fields. Trace element contents below the limit of detection are represented as half of the detection limit values of the LA-ICP-MS instrument on the

diagram. The data of pyrite (TAG-Py) and marcasite (TAG-Mrc) at the Trans-Atlantic Geotraverse (TAG) are sourced from [Grant et al. \(2018\)](#).

Fig. 9. Box plot (a) and histogram (b) illustrating the variation in $\delta^{34}\text{S}$ values of pyrite (Py), marcasite (Mrc), and chalcopyrite (Ccp) from the Xunmei (yellow) and Niaochao (red) vent fields. Panel (c) presents detailed $\delta^{34}\text{S}$ values of sulfides within the sulfide chimneys and fragments.

Fig. 10. Paragenetic sequence of mineralogical along with geochemical and isotopic signatures of sulfides at the Xunmei and Niaochao hydrothermal fields. Note: ‘-’ means not analyzed. The fluid temperature gradually increases from the outer to the inner walls of sulfide chimneys ([Graham et al., 1988](#); [Tivey, 1995](#)). Each mineral type has a specific temperature range, defined by key thresholds: anhydrite forms above 150°C, marcasite below 240°C, sphalerite between 259 and 299°C (mean: 282°C), and chalcopyrite from 300 to 350°C. These ranges align with the findings of [Dang et al. \(2023\)](#).

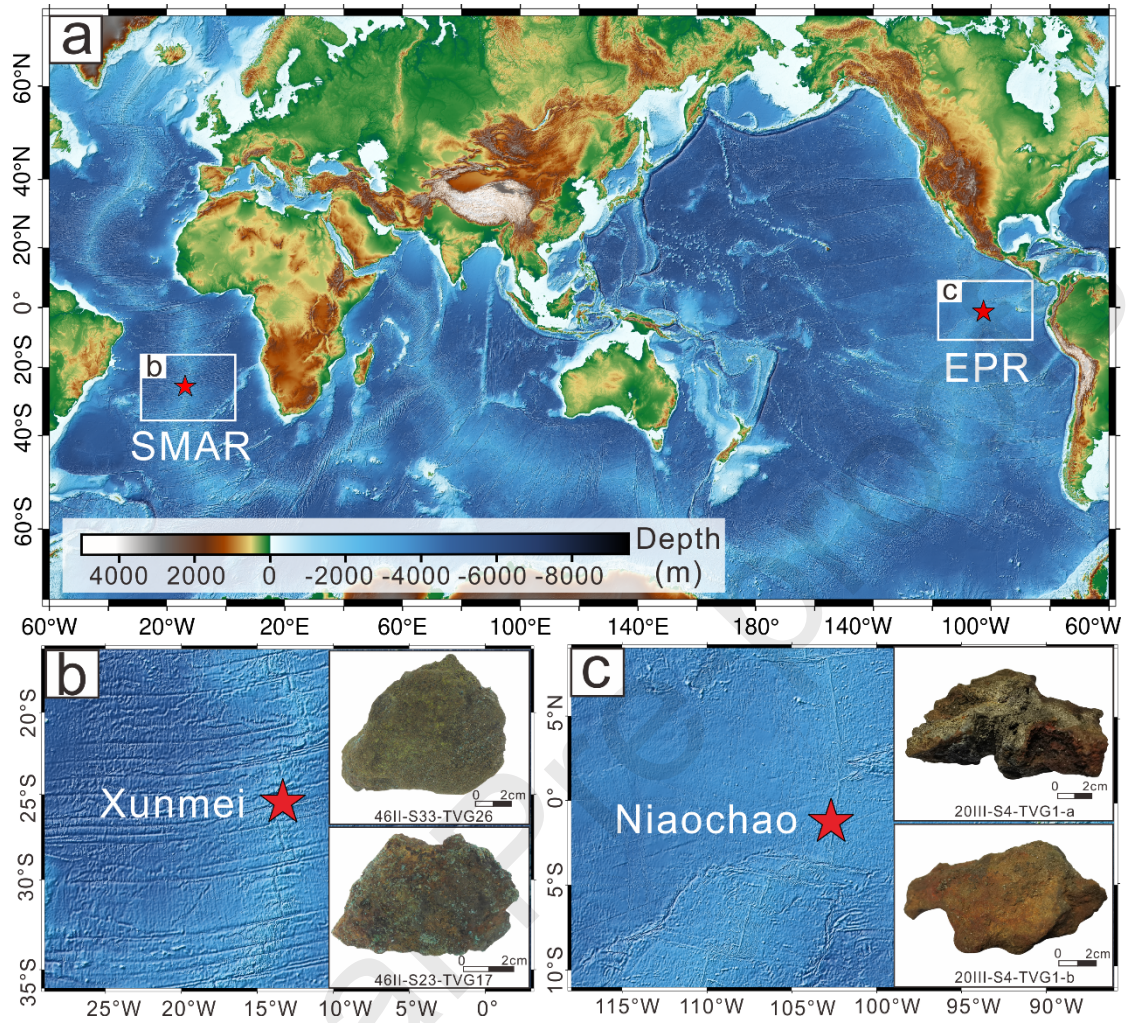
Fig. 11. Representative LA-ICP-MS time-resolved depth profiles for trace elements in marcasite (Mrc) and pyrite (Py) from the Xunmei (a-d, g, h) and Niaochao (e, f) vent fields. (a) The profiles of Zn, Ag, and Mo in marcasite (I-Mrc1) exhibit abrupt peaks, indicating the presence of inclusions of these elements. (b) In pyrite (I-Py1), profiles for Zn, Pb, Ag, Mo, and Mn reflect the occurrence of sphalerite, galena, Ag-bearing, Mo-bearing, and Mn-bearing inclusions, respectively. (c) In marcasite (I-Mrc2), zinc shows covariations with Ag, suggesting an association with sphalerite, while spiky spectra of Cu and Mn indicate the presence of chalcopyrite and Mn-bearing inclusions. (d) Wavy spectra for Cu, Zn, Pb, and Ag observed in pyrite (I-Py2) suggest the appearance of submicron inclusions beneath the mineral surface. (e) Spiky spectra of Cu, Pb, and Mn in marcasite (II-Mrc2) are indicative of chalcopyrite, galena, and Mn-bearing inclusions, respectively. (f) The spiky spectra of Cu, Zn, Pb, and Sb indicate that they occur as inclusions within pyrite (II-Py2). (g) In marcasite (I-Mrc3), Cu exhibits covariations with As and Ag, likely linked to the presence of tetrahedrite-tennantite; meanwhile, covariations between Mn and Mo suggest the occurrence of Mn-Mo-bearing inclusions. (h) Elements such as Cu, Zn, Pb, and Mo exhibit spiky spectra, signifying that these metals occur as inclusions in pyrite (I-Py3).

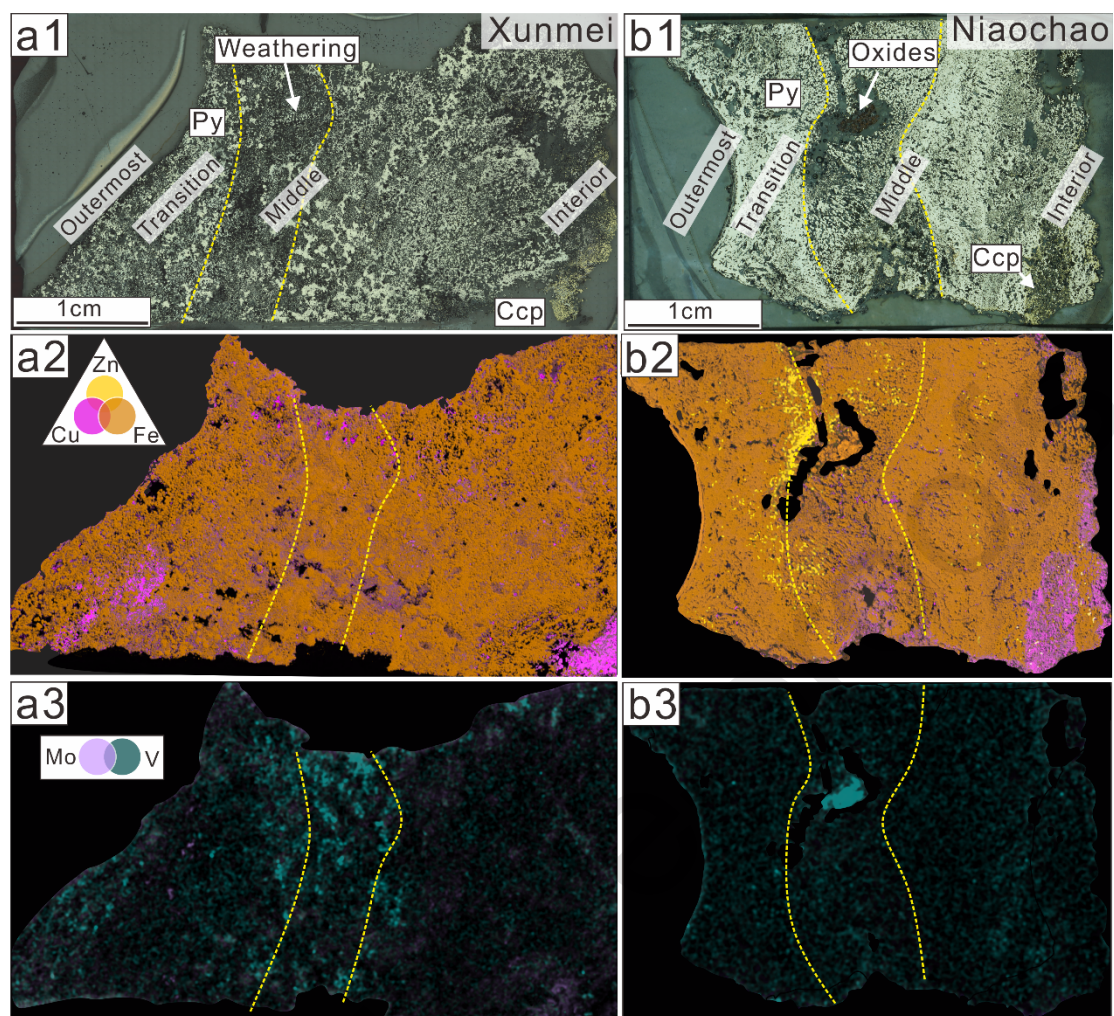
Fig. 12. Percentage of seawater-derived sulfur within sulfides across the Xunmei and Niaochao chimneys using a two-endmember mixing model ([One et al., 2007](#)).

Fig. 13. Seawater-derived metals V and Mo vs. $\delta^{34}\text{S}$ values of the pyrite (Py) and marcasite (Mrc) in the Xunmei chimneys. (a) V vs. $\delta^{34}\text{S}$, (b) Mo vs. $\delta^{34}\text{S}$. Data of I-Py3 in the grey region are outliers.

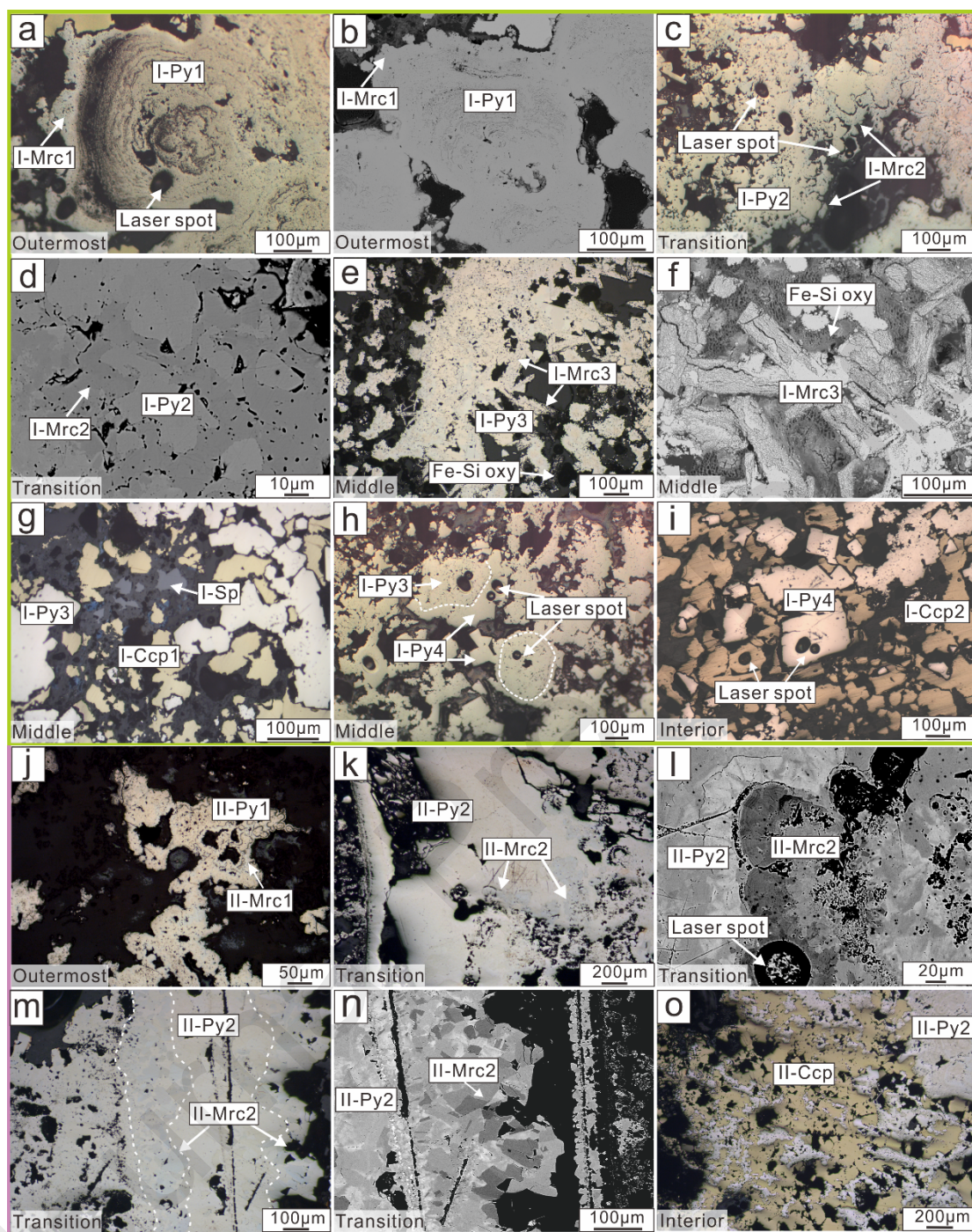
Fig. 14. The schematic model of the microchemical variations between marcasite and pyrite in sulfide chimneys. (a) General schematic zonation of a chimney (modified from [Wohlgemuth-Ueberwasser et al., 2015](#)). (b) Chemical variations within the colloform marcasite-pyrite intergrowths (I-Mrc1 and I-Py1) located in the outer chimney walls. Toward the transition zone, (c) the replacement and dissolution of pyrite (I-Py2) by marcasite (I-Mrc2) cause metal remobilization, (d) while marcasite-pyrite formed through the replacement of marcasite (II-Mrc2) by pyrite (II-Py2) shows metal depletion. (e) Marcasite pseudomorphs (I-Mrc3) after pyrrhotite were cemented by anhedral pyrite (I-Py3) as seawater infiltrated into the chimney. (f) Coarse-grained

1116 marcasite (Mrc), induced by fluid boiling, was enveloped by chalcopyrite (Ccp) (refer
 1117 to [Dang et al., 2023](#)).

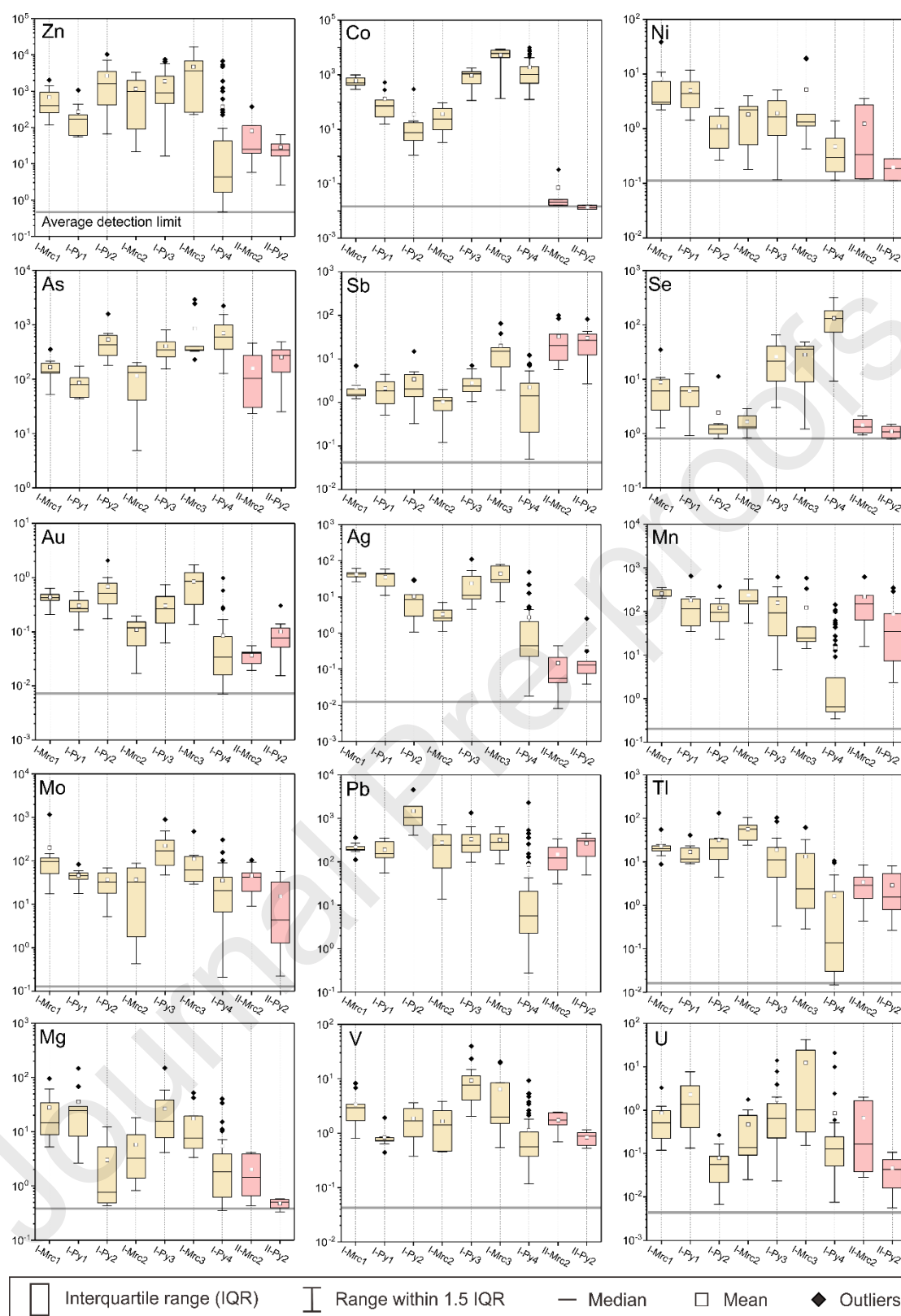


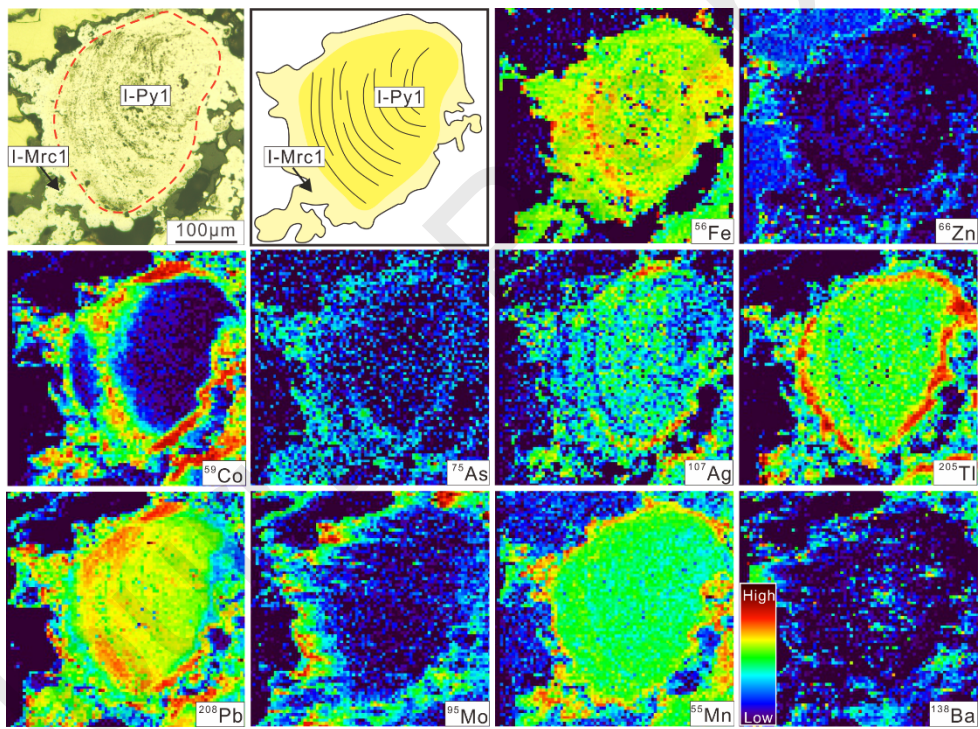
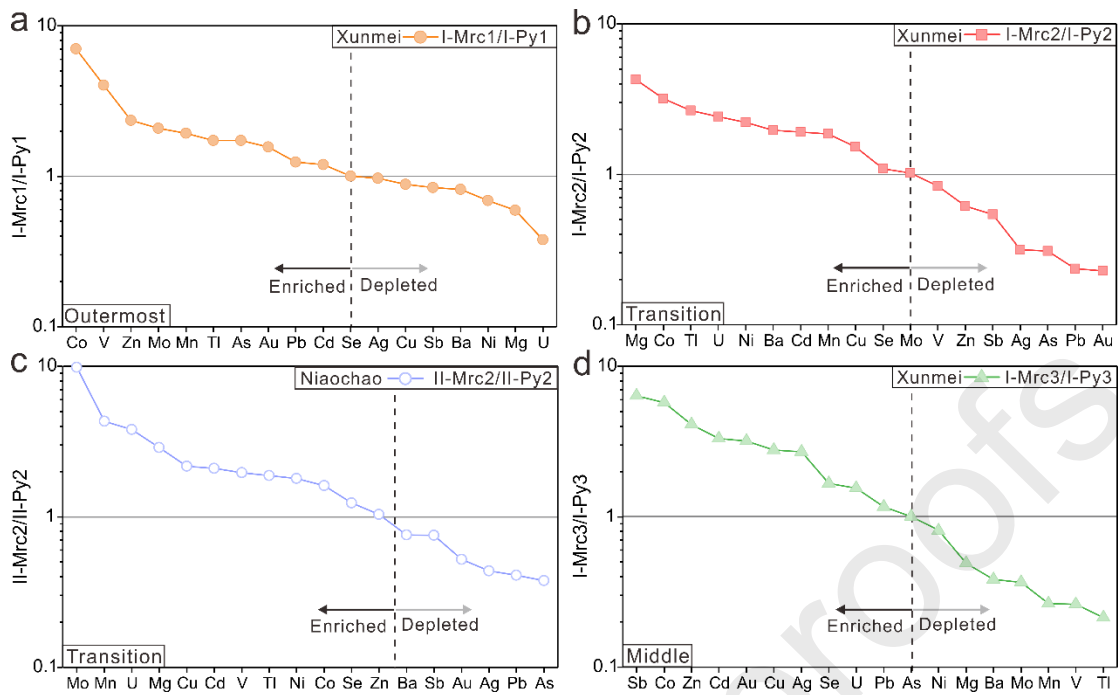


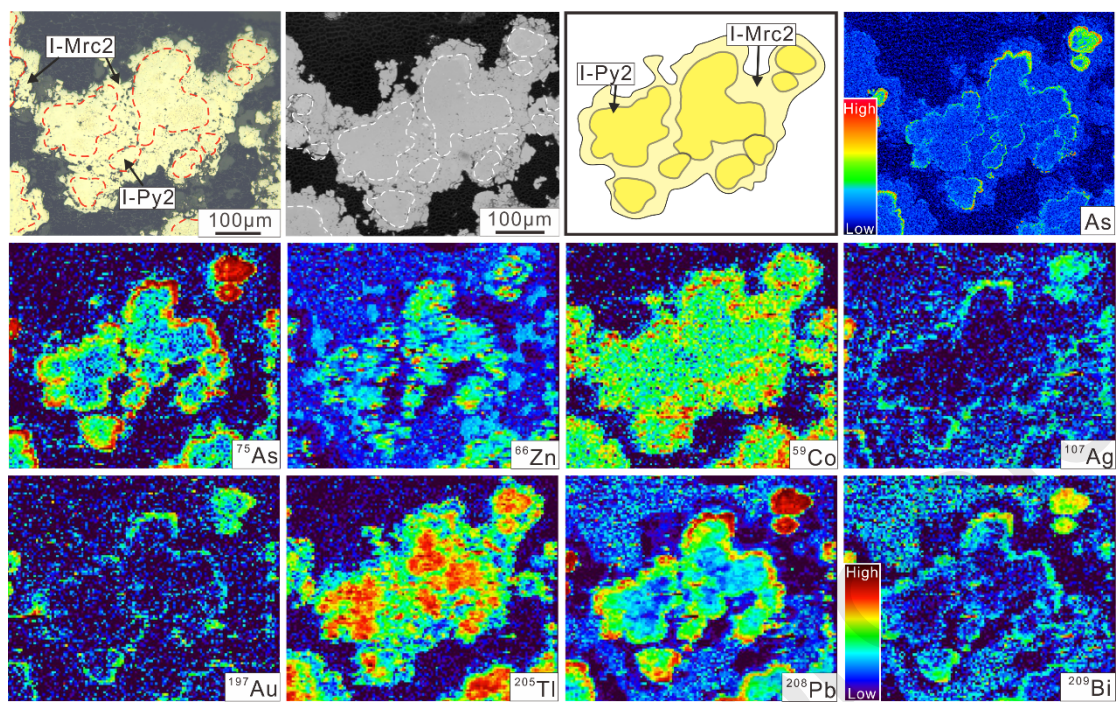
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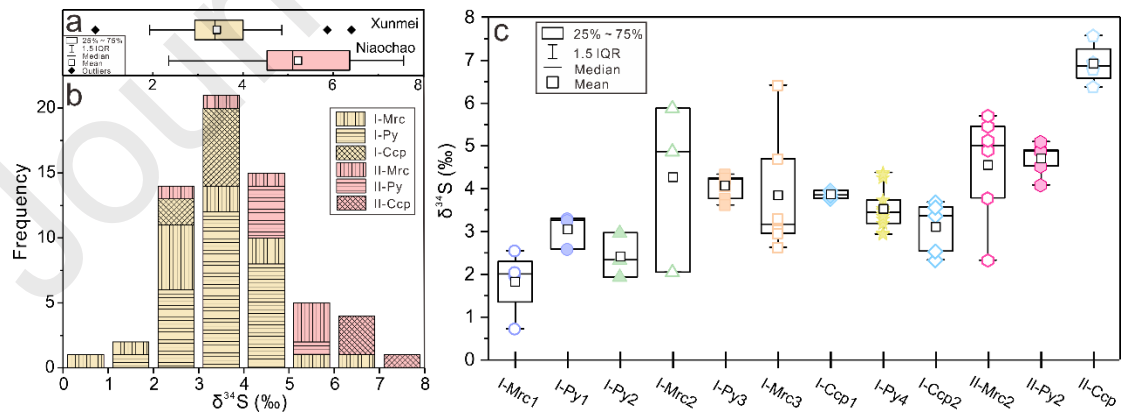
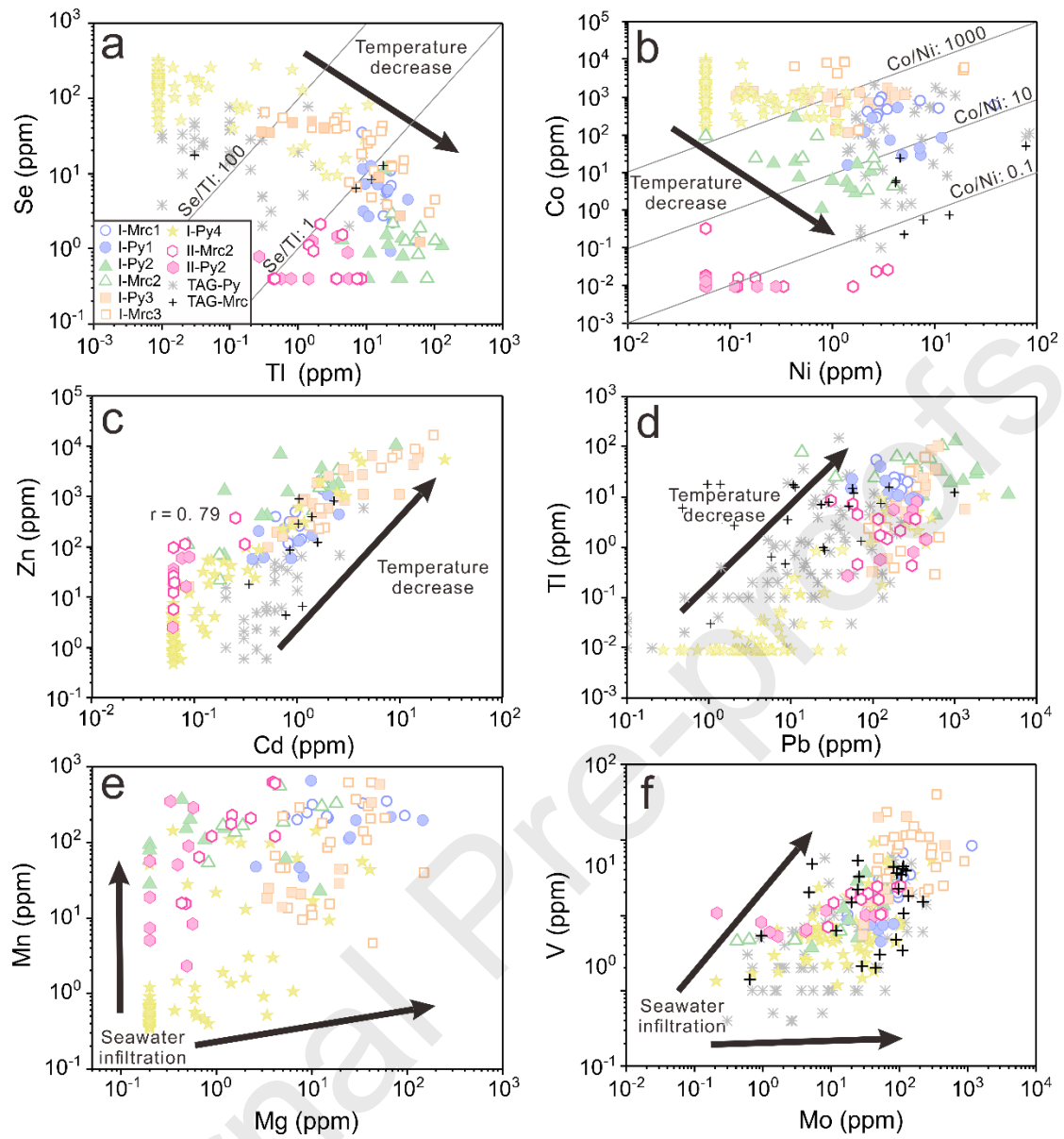
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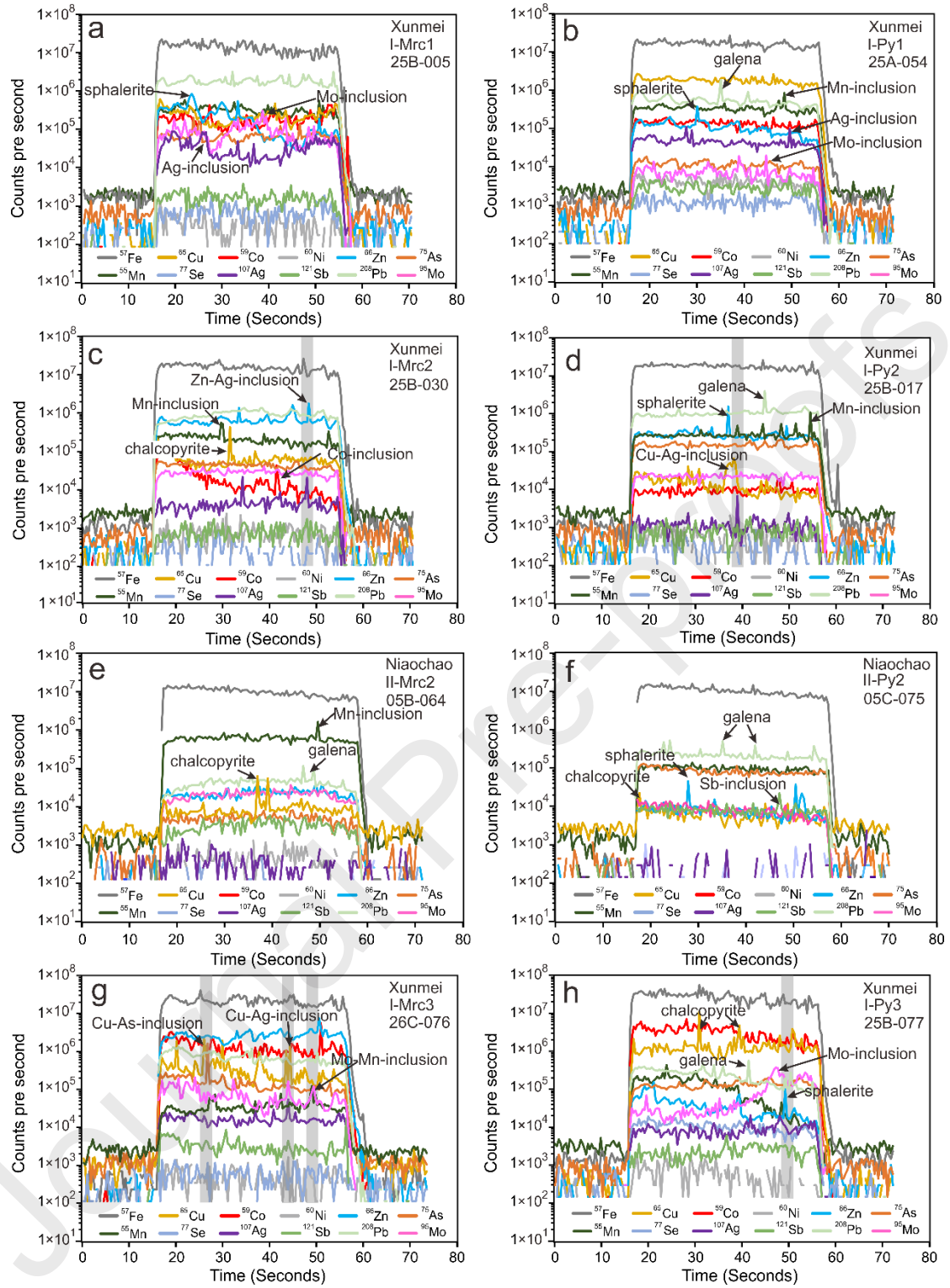


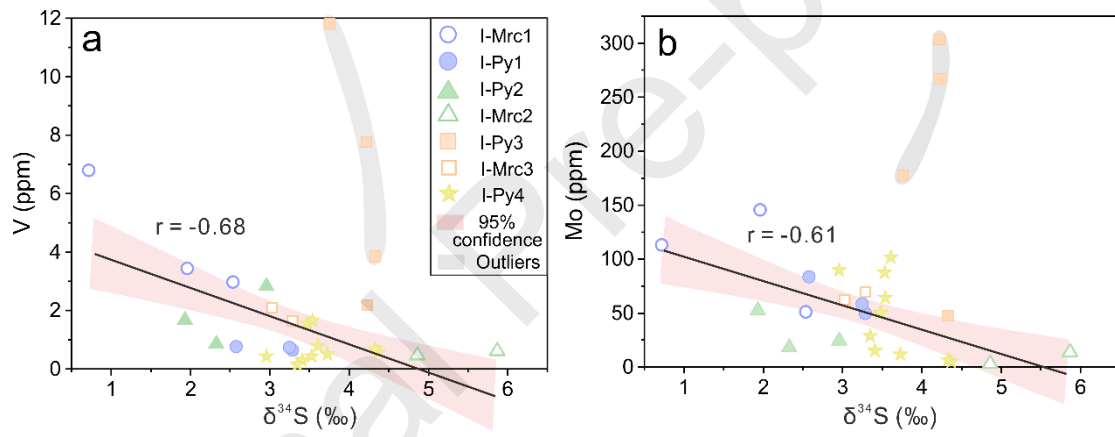
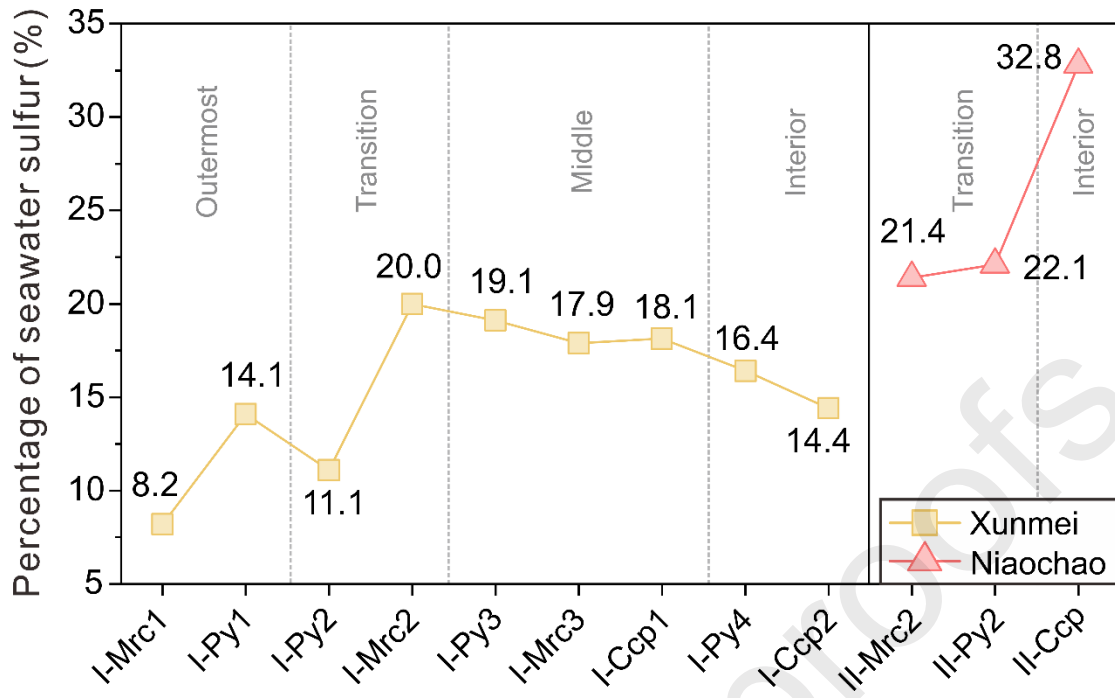
Temperature (°C)	Low-T		High-T		Type of Py-Mrc	Associations of trace elements	Mean $\delta^{34}\text{S}$ (‰)
	<150	>240	282	<350			
<i>Xunmei vent field</i>							
Anhydrite	■					—	—
I-Marcasite1	■				1	Co, V, Zn, Mo, Mn, Tl, As, Au, Pb	1.82
I-Pyrite1	■	■				U, Mg, Ni, Ba, Sb, Cu	3.04
I-Pyrite2		■			2	As, Au, Pb, Ag, Sb, Zn	2.41
I-Marcasite2		■				Mg, U, Co, Tl, Ba, Cd, Mn, Cu	4.26
I-Pyrite3			■	■	4	Tl, V, Mn, Mo, Ba, Mg	4.06
I-Marcasite3			■			Sb, Co, Zn, Cd, Au, Cu, Ag	3.83
I-Sphalerite				■		—	—
I-Chalcopyrite1				■		Se, In, Ag	3.86
I-Pyrite4				■		Co, As, Se	3.52
I-Chalcopyrite2				■		Se, In, Ag	3.09
<i>Niaochao vent field</i>							
II-Pyrite1	■				1	—	—
II-Marcasite1	■					—	—
II-Marcasite2		■			3	Mn, Mg, Mo, Tl	4.54
II-Pyrite2			■	■		As, Sb, Pb, Au	4.69
II-Chalcopyrite				■		Se, In, Sn	6.92

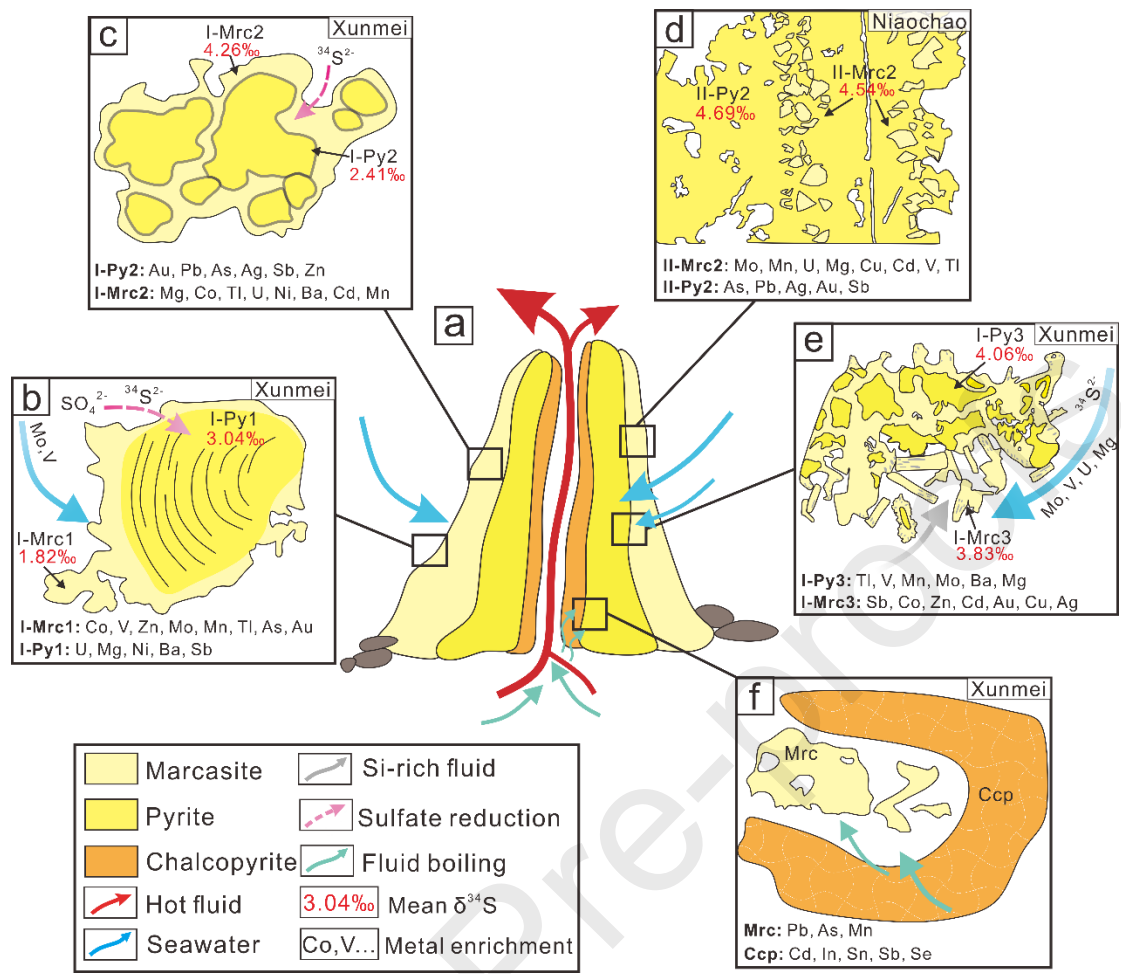
■, abundant; ■, common; ■, minor.

■, abundant; ■, common; ■, minor.

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